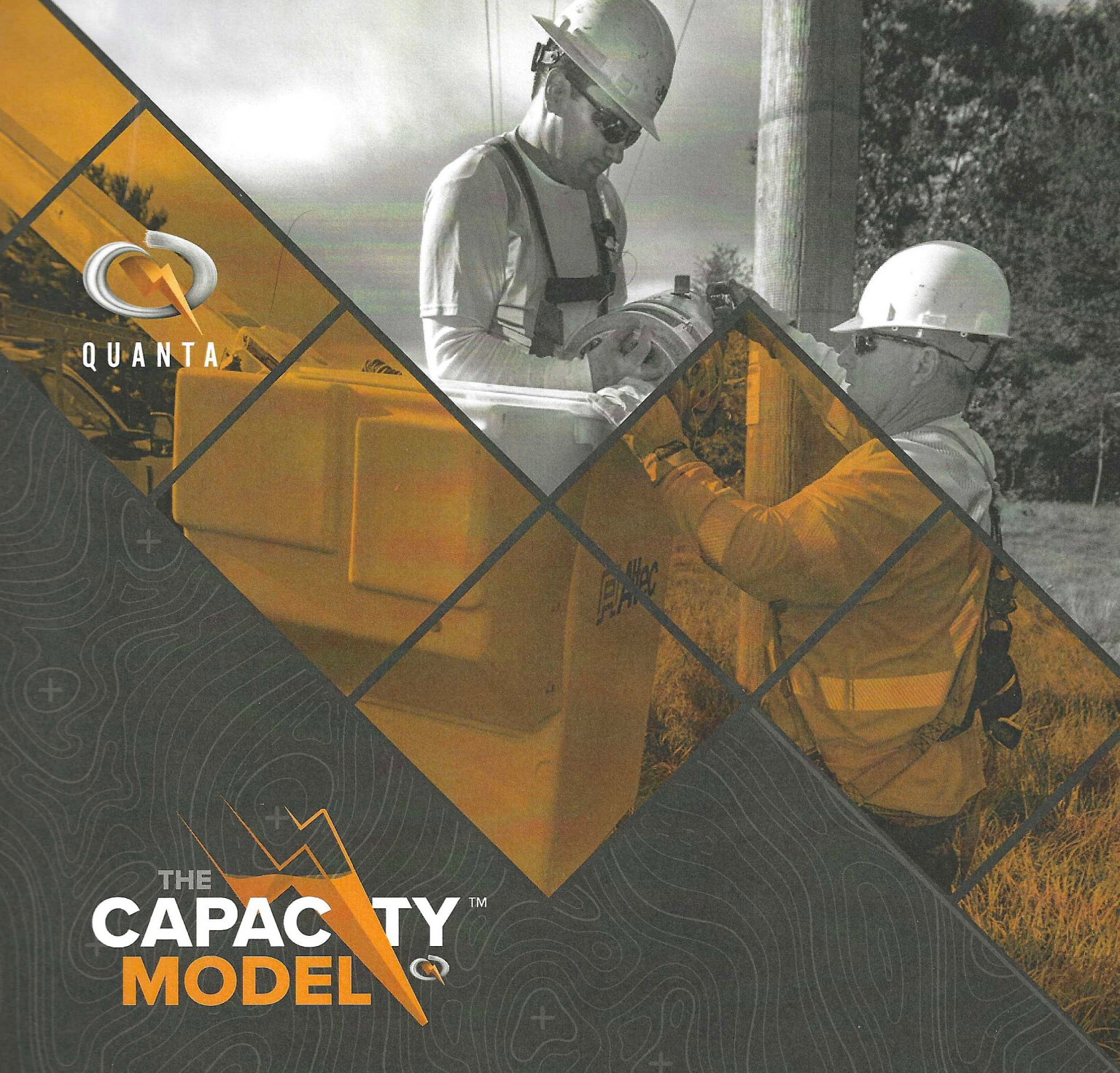


QUANTA

THE CAPACITY MODEL™

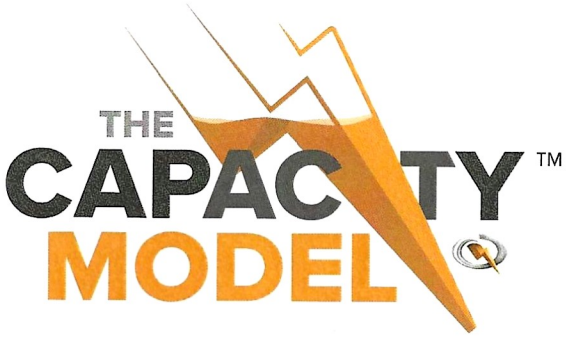
QUANTA HUMAN PERFORMANCE PRINCIPLES

FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

INSTRUCTOR MANUAL

V 1.0

NLC


THE
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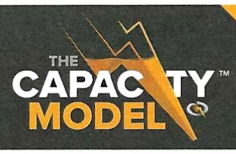

QUANTA HUMAN PERFORMANCE PRINCIPLES

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Quanta Human Performance Principles for the Power and Telecommunications Industry

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TCM 3001

HUMAN PERFORMANCE PRINCIPLES FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

SYLLABUS

COURSE DESCRIPTION

Trainees learn how the Quanta Human Performance Principles support The Capacity Model®, how to apply the learning approach to accountability, and act on events with a productive response. Trainees who complete the course taught by an NLC instructor will receive a certificate of completion from NLC.

The class is taken in conjunction with the STKY and the Energy Wheel module. Complete both modules within three months to gain maximum learning and retention benefits.

Intended audience: Operations management, safety professionals, supervisors who direct the work of others, and those responsible for leading job briefings.

LEARNING OUTCOMES

1.a. Explain what The Capacity Model® is.
1.b. Explain how the Quanta Human Performance Principles support The Capacity Model®.
1.c. Compare the learning and blameapproaches to accountability.
1.d. Select a productive response for a given outcome.

SUGGESTED PREREQUISITES

- Understanding The Capacity Model®

COURSE MATERIALS

- Materials to be issued at the beginning of the course.

COURSE OUTLINE

- Housekeeping
- Quanta Human Performance Principles
- Exam

INSTRUCTIONAL METHODS

- Lecture
- Learning activities
- Knowledge check

METHODS OF MEASUREMENT

KNOWLEDGE
Written exam:
A = 100%–90%
B = 89%–80%
C = 79%–70%
D = 69%–60%
F = 59% and below

If a student earns less than a 70% on the exam, one retake is permitted. If they do not earn at least a 70% on the exam retake, they must retake the course.

SKILL
Completion of all activities as directed by the instructor.

BEHAVIOR
Active participation and engagement in class discussions and activities.

STUDENT-TO-INSTRUCTOR RATIO
Maximum of 24:1.

CLOCK HOURS
3 Hours

POLICIES AND EXPECTATIONS

Building a positive learning environment—Be sure to share, support your classmates, and contribute to the positive learning environment during your course. You bring incredible value to the group and to the course. Please refrain from any behavior that you know doesn't work in a classroom environment.

TCM 3001
HUMAN PERFORMANCE PRINCIPLES FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

AGENDA

SCHEDULE

HOUSEKEEPING	*15 minutes*
QUANTA HUMAN PERFORMANCE PRINCIPLES	*2 hours 15 minutes*
EXAM	*30 minutes*

Breaks are scheduled at the discretion of the instructor.

The lunch schedule is determined by the class start time and the option to complete the STKY and the Energy Wheel module in the same day.

USER GUIDE

INTRODUCTION

The slides and notes presented in this manual may look slightly different than those in the PowerPoint. They include additional instructions and support to help you teach the course. The slides images include symbols that align with symbols in the corresponding notes to help you quickly see what needs to be done during class. See Table 1.1 for a list of the symbols and their meaning.

Read through this manual before teaching the course, and have the notes with you during the lesson. Use this manual to record your training notes, application activities, stories, and so forth alongside the appropriate notes in this manual.

SYMBOLS

SYMBOL	MEANING
	Stop here during the presentation to ask the corresponding discussion question.
	Tell a story about the topic indicated in the instructor notes.
	Perform a formative assessment to guide your instruction.
	Complete one or more of the provided application activities.
	Look here for a helpful teaching tip.
	This is a good place to review the guiding question. There is no need for discussion. Simply remind the students of the guiding question.

Table 1.1 The symbols are used on the slide images and corresponding notes to provide additional guidance for teaching the course.

MODULE OVERVIEW

GUIDING QUESTIONS

The guiding question is a focus of the lesson. Present it to the trainees at the beginning of the course and reinforce it throughout the training. Repetition is key in learning and this simple question reinforces the importance of the topic.

- How can our responses to an outcome effect change?

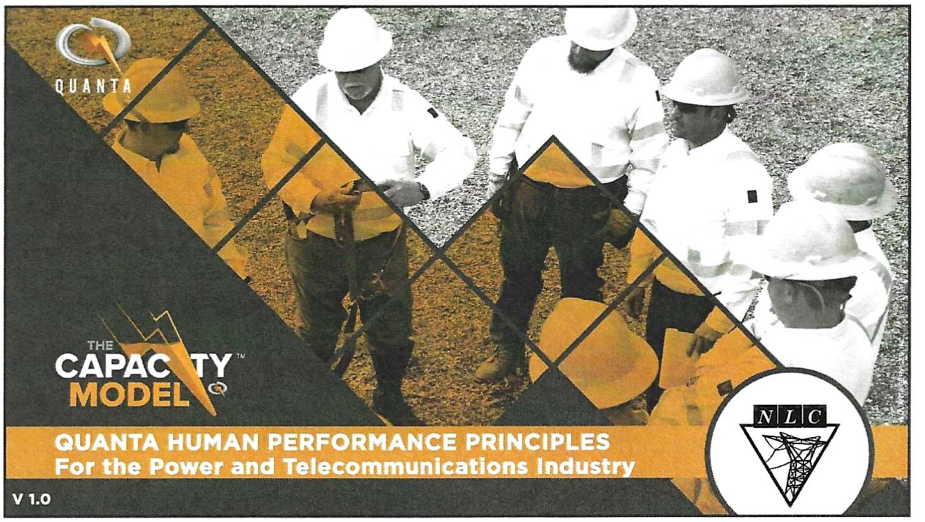
KEY TERMS

accountability	An obligation to accept responsibility and account for or explain one's actions.
capacity	The actual or potential ability to perform, yield, or withstand. With The Capacity Model, the ability to perform work safely, yield safe results, and withstand the outcome of an accident or mistake.
capacity for failure	In a work setting, the ability to perform a task while safely withstanding the outcome of an accident or mistake.
human performance	In the context of worker safety, the actual ways that people interact with their real-life work environment—their tasks, equipment, coworkers, physical setting, and culture. It accounts for how normal human behavior plays out in the workplace.
operational learning	An approach to learning about the context of an event to promote successful operations and prevent negative events from repeating. For example, instead of firing someone who made a mistake, find out the circumstances that caused that person to make the mistake—and then fix those circumstances.
mitigative controls	Acts, objects, or systems designed to reduce the consequences of an unwanted event. Mitigative controls help after the bad stuff has already happened.
preventive controls	Acts, objects, or systems designed to prevent an unwanted event. Preventive controls stop the bad stuff from happening.
reaction	An immediate, unconscious behavior driven by an emotional trigger, such as anger or surprise.
response	An intentional action based on a conscious decision.

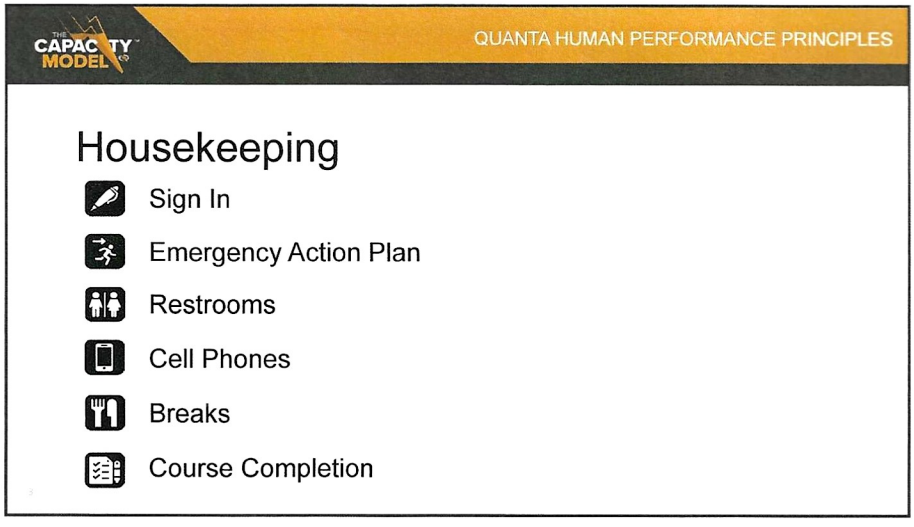
POWERPOINT PRESENTATION GUIDE

Use the corresponding notes below each slide image for additional guidance on delivering the content.

Show this slide while trainees are arriving.

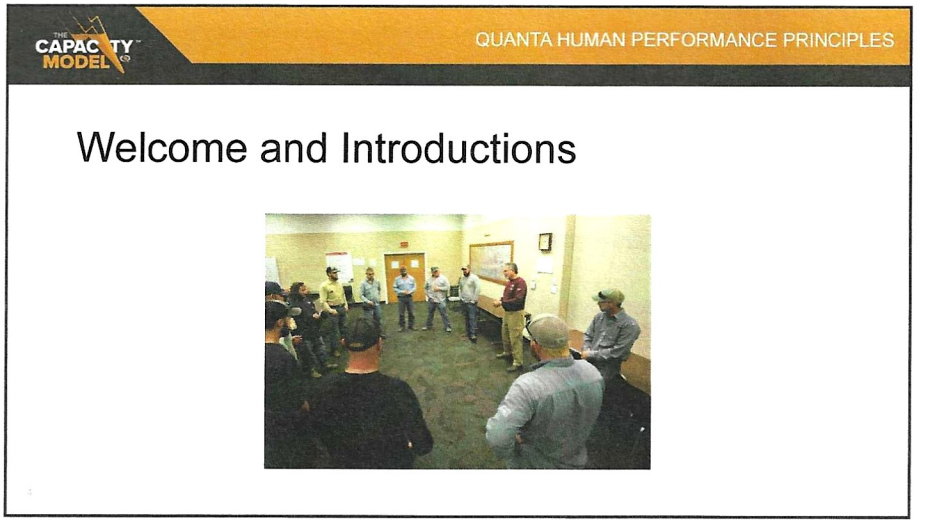


Briefly introduce yourself and welcome the trainees to the training.



1. Pass around a sign-in sheet and ensure that all trainees sign in.
2. Complete the emergency action plan:
   - Point out the emergency exits.
   - Explain where to meet up in case of an emergency.
   - Ask trainees who are trained in first aid and CPR to identify themselves. Choose one to be responsible for first aid and CPR and an alternate, if possible, in case the primary person is involved in the emergency.
   - Explain the location of the nearest AED.
   - Provide your cell number in case of an emergency.
   - Post the address of the training location in the room.
   - Identify one person to meet EMS in an emergency and an alternate in case the primary person assigned is involved in the emergency.
3. Provide the location of the nearest restrooms.
4. Explain the cell phone policy: Put on silent, take necessary calls outside the classroom.
5. Explain breaks and lunch (if a lunch break is applicable)..
   - You can give breaks anytime you think trainees need one.
   - A good rule of thumb is to provide a 5-10 minute break at least every hour or two, depending on the level of engagement and the pace.
   - Look for signs of disengagement, slouching posture, etc. and provide breaks accordingly.
6. Explain Human Performance Principles is one module of the course. The other module is STKY and the Energy Wheel.
   - To complete this module, trainees must participate in the class and earn at least a 70% on the exam. Explain the exam is 10 multiple-choice questions.

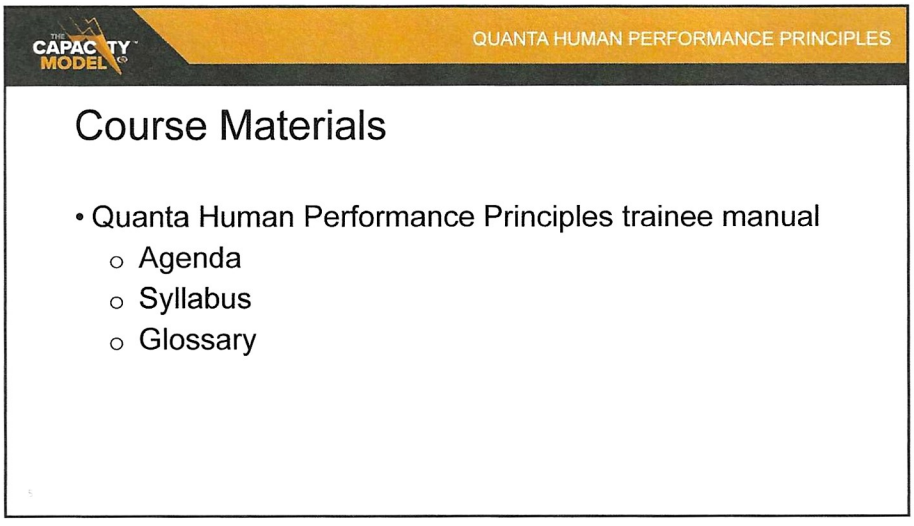
Identify a student who will advise you when the class may need a break.



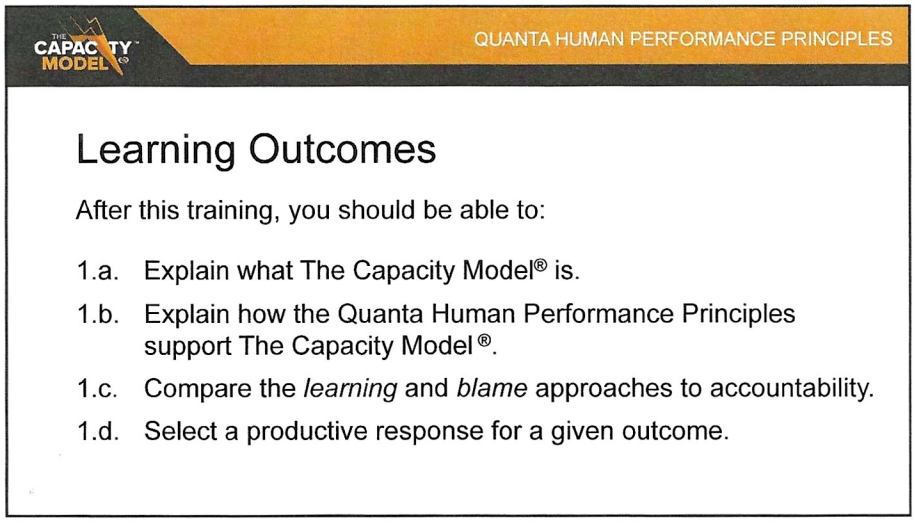
- Introduce yourself and share your background.
- Ask trainees to introduce themselves and share their backgrounds.
- Use this beginning segment to build rapport and establish credibility (positive learning environment).

Consider these questions to help with introductions.

- Who had the greatest influence in your career?
- Where was the most interesting place you have worked?
- What quality makes you good at your job?
- What does it mean to be safe?

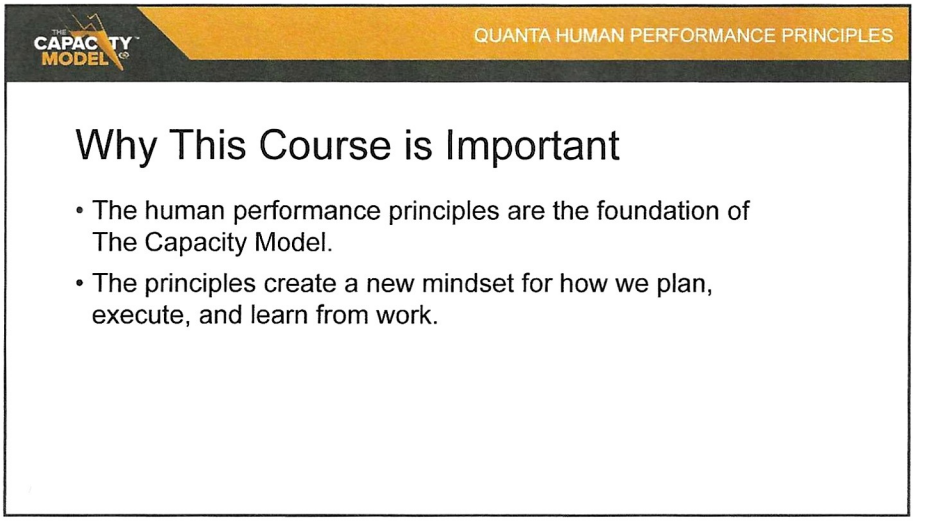


- Confirm trainees received a manual.
- Point out the syllabus and agenda on pages 4 and 5 of the manual.
- Point out the key terms in the manual and PowerPoint are also in the glossary on page 35.
- Explain the manual aligns with the PowerPoint presentation.

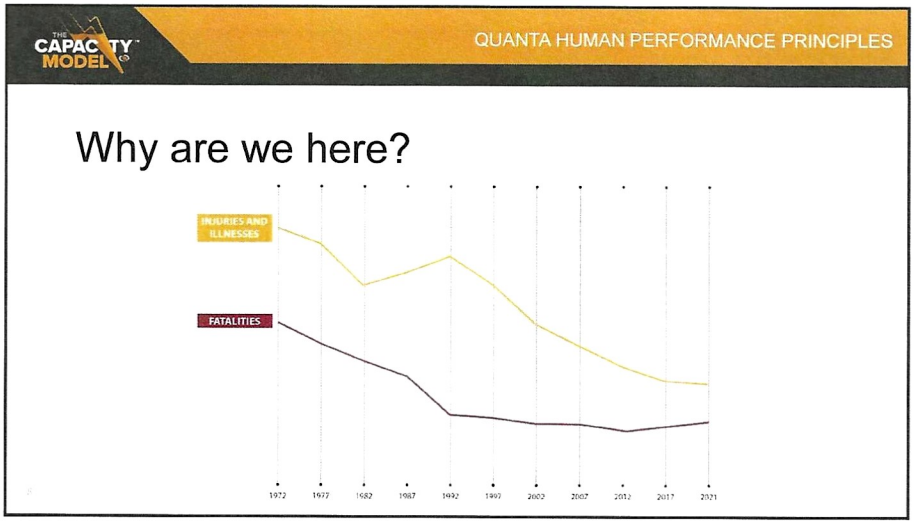


Slide aligns with page 7 of the trainee manual.

- Review learning outcomes.
- Read or ask for volunteers to read the learning outcomes.
- Write the learning outcomes on a whiteboard or post them in a visible area of the room to refer to during the training.



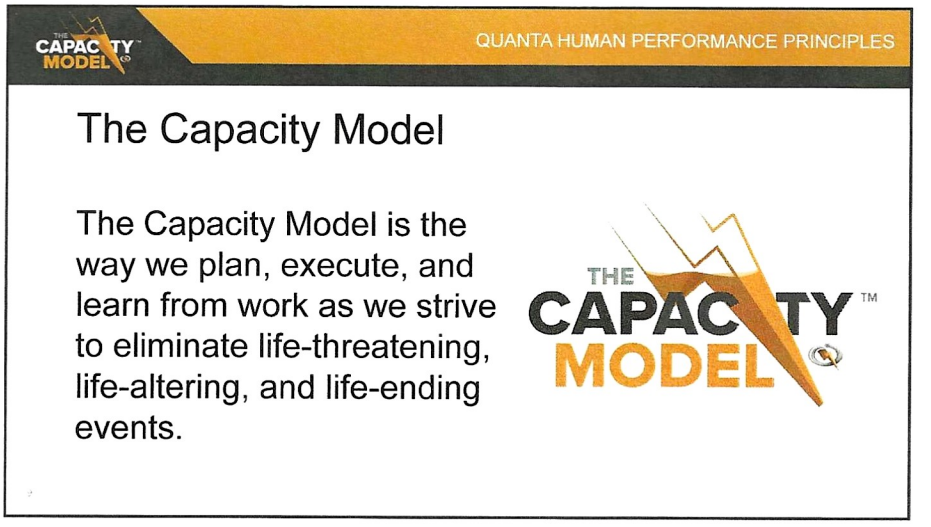
Use the bullet items to discuss why this course is important for the trainees.



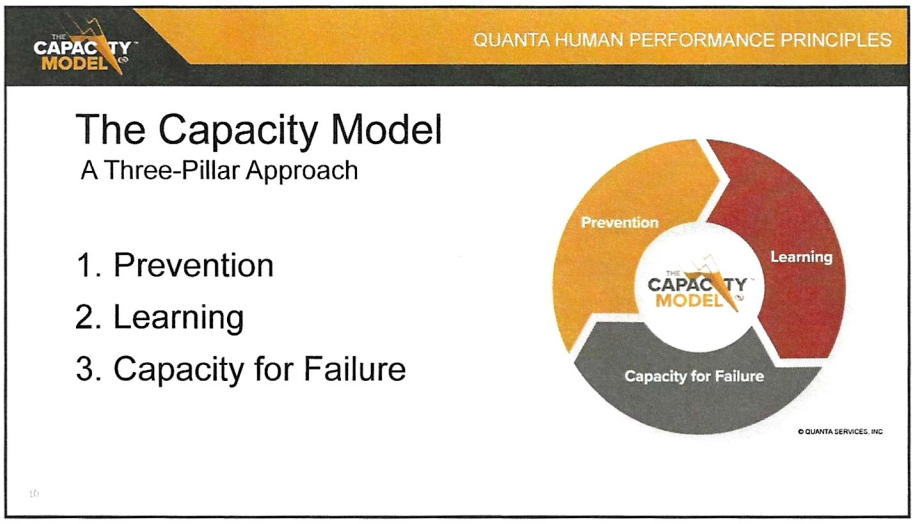
- Direct trainees to page 10 in their manual to review the statistics on the slide.
- Discuss how the Occupational Safety and Health Act of 1970 and the creation of the Occupational Safety and Health Administration (OSHA) in 1971 were key in reducing the number of injuries and fatalities.
- Point out how the fatal injury rate declined for many years before reaching a low of 3.3 per 100,000 full-time equivalent workers in 2013, before leveling out and then increasing to 3.6 per 100,000 full-time equivalent workers in 2021.

PRESENT THE FOLLOWING:

- Traditional safety has focused primarily on preventing accidents and incidents. The theory is that if we prevent these unsafe acts from occurring, there should be an overall reduction in incidents, including fatalities.
- Instead, prevention has been proven to have a profound effect on reducing near misses and minor injuries but a limited effect on reducing major injuries and fatalities.
- Workers must also focus on events that are life-threatening, life-altering, or life-ending.
- Let's review The Capacity Model—Quanta's safety philosophy—and the three pillars of The Capacity Model.



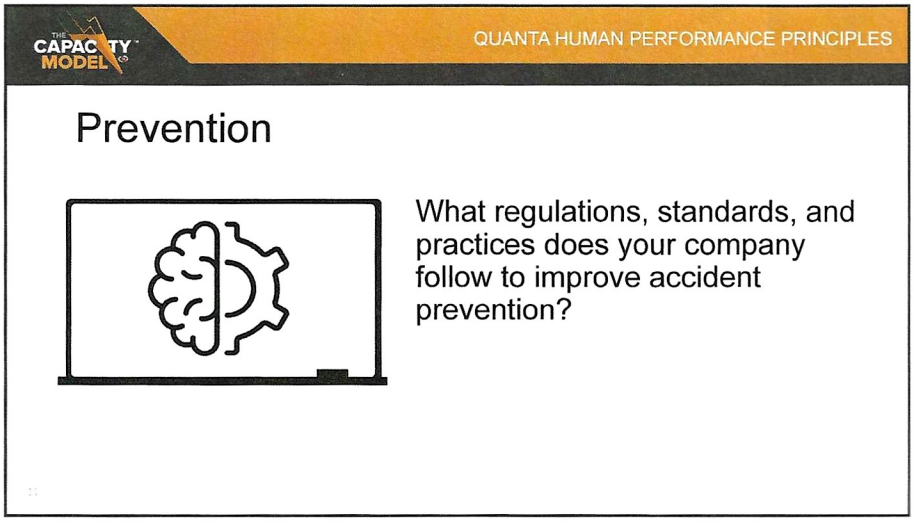
- Read the definition on the slide.
- Break down each element of the definition. For example:
- Eliminate life-threatening, life-altering, and life-ending events: While we always want to eliminate and mitigate all events that can cause injuries, The Capacity Model focuses on STKY events – the stuff that kills you.
- Plan work: Everything we do before the work begins allows us to focus on the tasks we perform and the preventive controls that are necessary.
- Execute work: But remember, prevention is not enough. As people execute the work in real time, they are continuously adapting and adjusting to normal conditions and variability of the job. We know that sometimes, things do go wrong, so we will always plan our work as if failure will happen today and build in the capacity to fail safely when it does.
- Learn from work: Whether learning from when things go wrong, or from things that go right, we know that learning is a deliberate improvement strategy. The only way to get better is to take the time to learn from those who do the work.



Slide aligns with page 11 of the trainee manual.

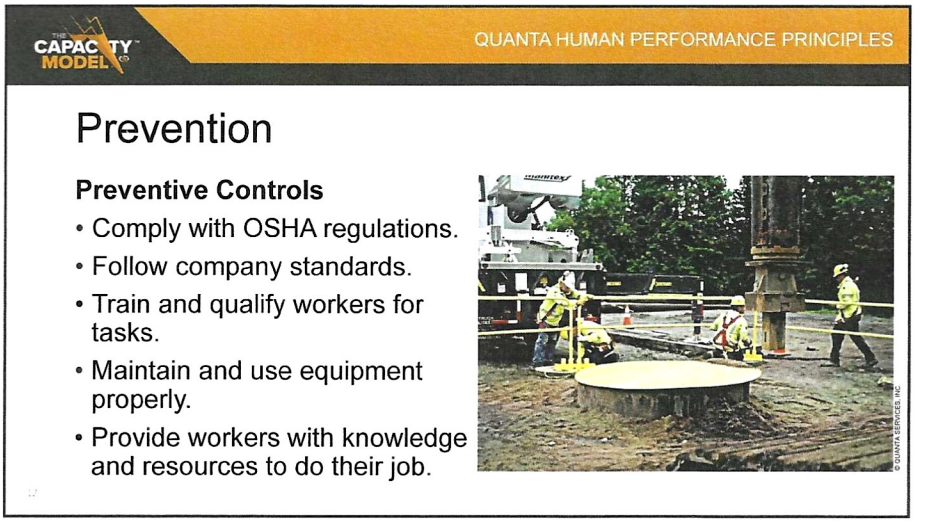
PRESENT THE FOLLOWING:

- The Capacity Model is a three-pillar approach to safety: prevention, learning, and capacity for failure.
- **Prevention:** Prevention is the traditional safety defense, and it has worked well. But there is still work we need to do to ensure everyone gets home safely every day; after all, People Make Mistakes (Principle 1).
- **Learning:** By creating an environment where we seek to learn from errors and incidents, instead of punishing our people, we can create trusting workplaces and uncover the true conditions that surround errors. We must view learning as a Deliberate Improvement Strategy (Principle 7).
- **Capacity for Failure:** Knowing that errors and failures are always around us, we must build resilience toward error with capacity. We must build capacity to safely absorb errors by implementing controls so that no one is injured when energy is released.
- We'll discuss each in more detail on the following slides.



ACTIVITY:

- Read the question aloud.
- Give trainees 1 minute to form a response.
- Ask for 1-2 volunteers to share their responses.
- Validate answers.



Slide aligns with page 12 of the trainee manual.

Review and discuss some of the traditional methods for accident prevention on the slide.

PRESENT THE FOLLOWING:

- Traditional safety measures like these focus primarily on prevention, which has worked well in reducing minor injuries like sprains, cuts, minor burns, and so forth.
- The regulations and standards in place today have improved workplace safety.
- We must continue to practice accident prevention through these measures.

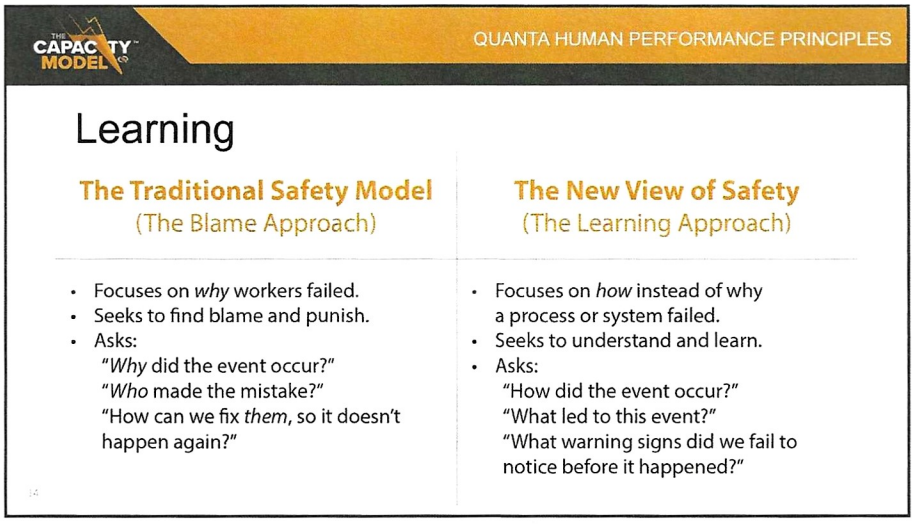
Review and discuss some of the preventive controls in the picture.

Slide aligns with page 12 of the trainee manual.

PRESENT THE FOLLOWING:

- We must also plan for failure because expecting perfection through prevention is not attainable. People make mistakes.
- Because people make mistakes, we must increase our focus on building capacity through mitigative controls. When a failure does occur, we can absorb that failure safely.

Explain the photo is taken at the same job site as the previous slide.
Review and discuss the mitigative control in the photo.



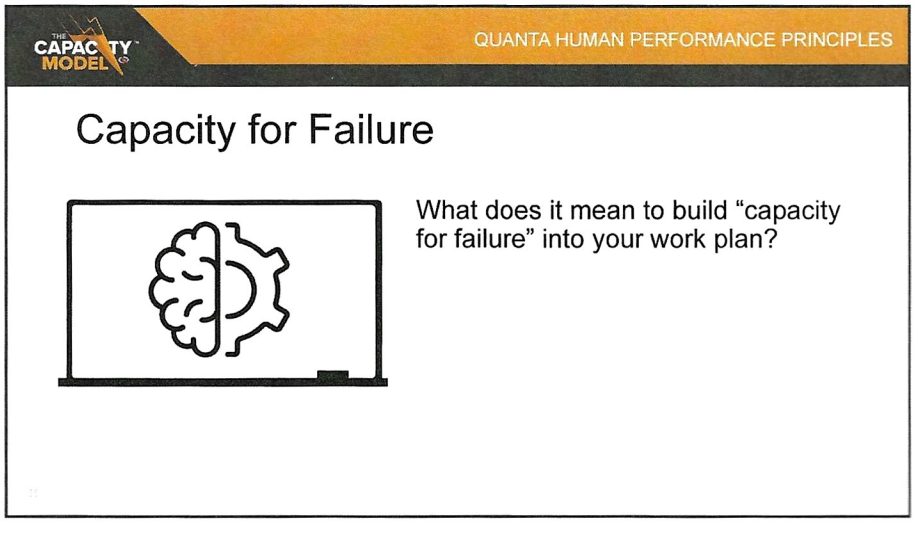
- Direct trainees to page 13 in their manual to review the figure on the slide.
- Review the elements of the traditional safety model.
- Review the elements of the new view of safety.

ACTIVITY:

Compare and contrast the two models as a class. For example:

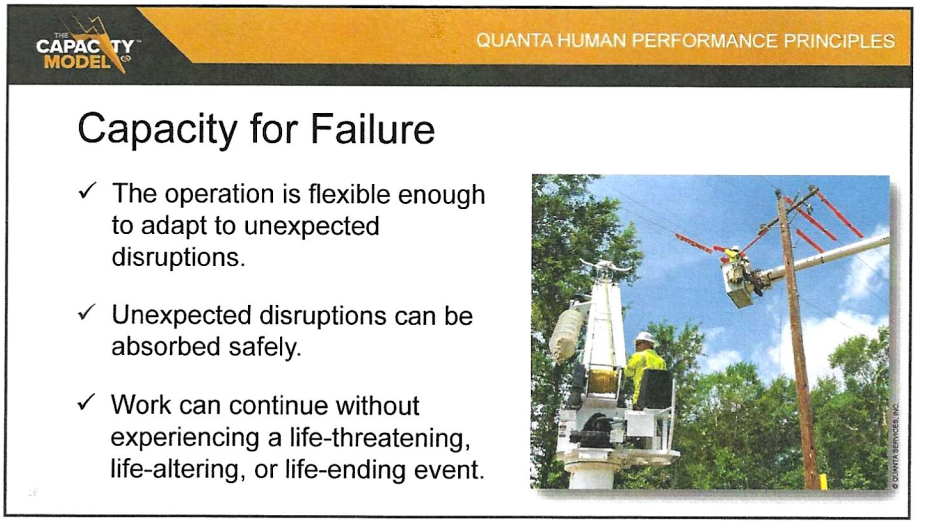
- What is the focus of the traditional safety model?
- What is the focus of the new view of safety?
- How do they differ?
- How does the traditional safety model prevent everyone from learning from incidents?

Introduce the guiding question:
How can our responses to an outcome effect change?



ACTIVITY:

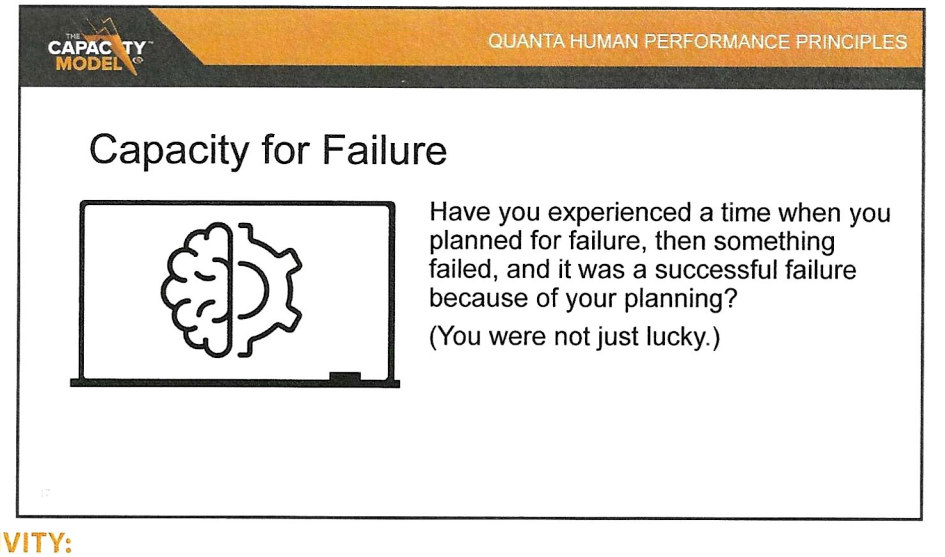
- Read the question aloud.
- Give trainees 1 minute to form a response.
- Ask for 1-2 volunteers to share their responses.
- Validate answers.



Slide aligns with page 14 of the trainee manual.

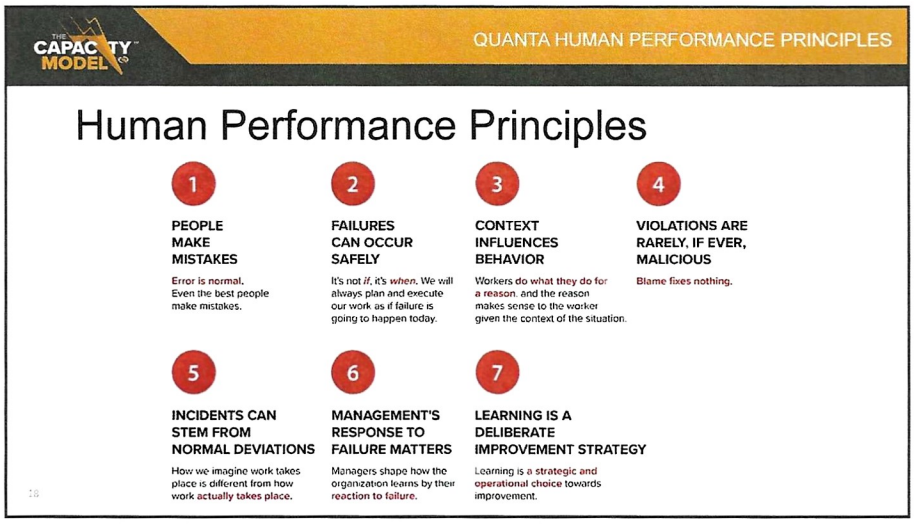
- Recall that mistakes and errors will happen on a job site, even with preventive controls in place.
- Explain that building capacity for failure are those mitigative controls they must put in place to allow those errors to occur safely.
- Use a scenario to walk through each of the items on the slide. For example:
  - During a critical lift, you have a marked exclusion zone.
  - When the load drops, the marked exclusion zone allows you to absorb that failure.
  - Because no one experienced a life-threatening, life-altering, or life-ending event.

- Discuss how to build capacity for failure:
  - Assume an accident or mistake will happen.
  - Identify where an accident or mistake could happen.
  - Ask, "How are we protected when that happens?"



ACTIVITY:

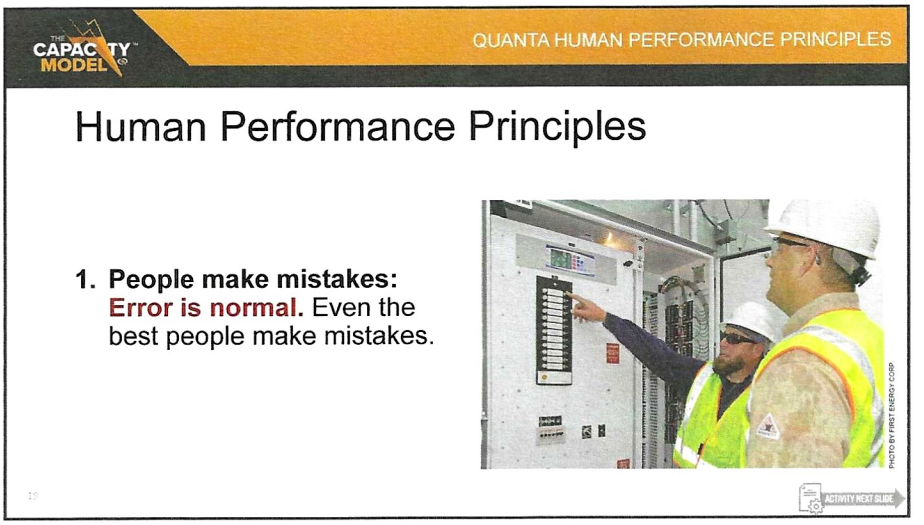
- Read the question aloud. If trainees do not have a personal story, ask them if they know a story of a successful failure.
- Share a personal experience as an example.
- Give trainees 1-2 minutes to form a response.
- Ask for volunteers to share their responses.
- Validate answers.



Slide aligns with page 16 of the trainee manual.

PRESENT THE FOLLOWING:

- Human performance is a way to think about how normal human behavior plays out in the workplace.
- As imperfect people deal with the constantly changing demands of their jobs, there will inevitably be failures and mistakes.
- No amount of training will ever change that.
- Managing a job site with this in mind helps both workers and supervisors find methods for allowing these mistakes to happen safely.
- Human performance is fundamental to The Capacity Model® and to improving safety. It's all about a mindset based on the capacity to fail safely.
- Quanta recognizes seven key human performance principles. We will discuss each in the following slides.

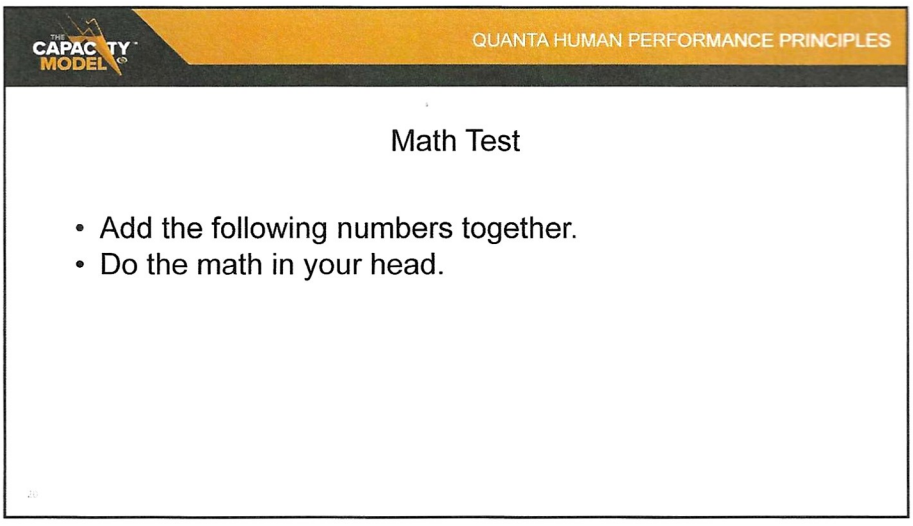


Slide aligns with page 17 of the trainee manual.

- Read #1 aloud.
- Share a mistake you made recently. It can be significant or insignificant.
- Ask if anyone in the group is willing to share a mistake they have made. Something they didn't intend to do.
- Give trainees 1-2 minutes to think about their responses.
- Call on 2-3 volunteers to share.

PRESENT THE FOLLOWING:

- Humans make mistakes all the time. When you watch the news, drive on roads with other drivers, or work with others, you can see that human error and failure are common.
- We use machines for specific tasks because they are great for repetitive, monotonous tasks. Think about assembly lines where machines do a lot of the work.
- More complex work requires the human interface.
- This is where we draw on our emotions, senses, experience, adaptation, and creativity to resolve complicated situations quickly.
- The part of the brain that governs these functions is separate from the part that governs things like logic. Because of this, absolute consistency in areas of work that require more complex thought is impossible.
- Mistakes are guaranteed, and errors should be treated as a source for learning and growth.

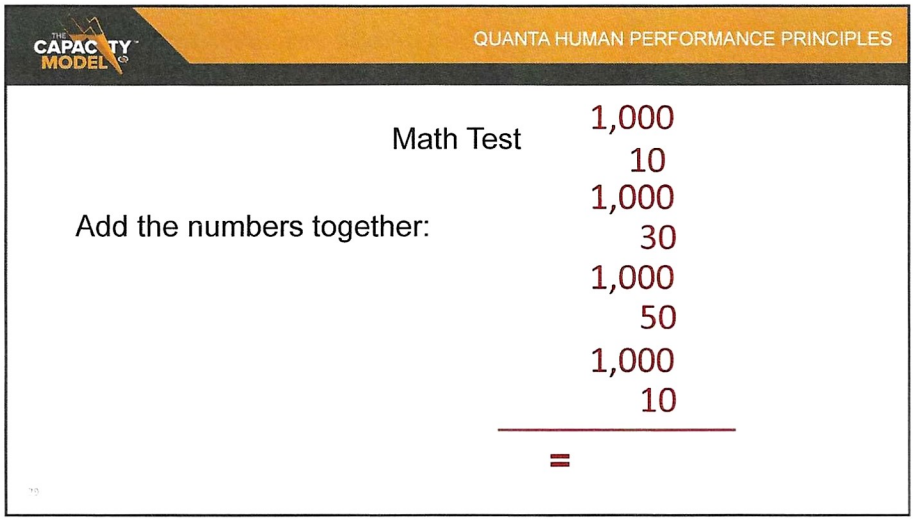


ACTIVITY:

PRESENT THE FOLLOWING:

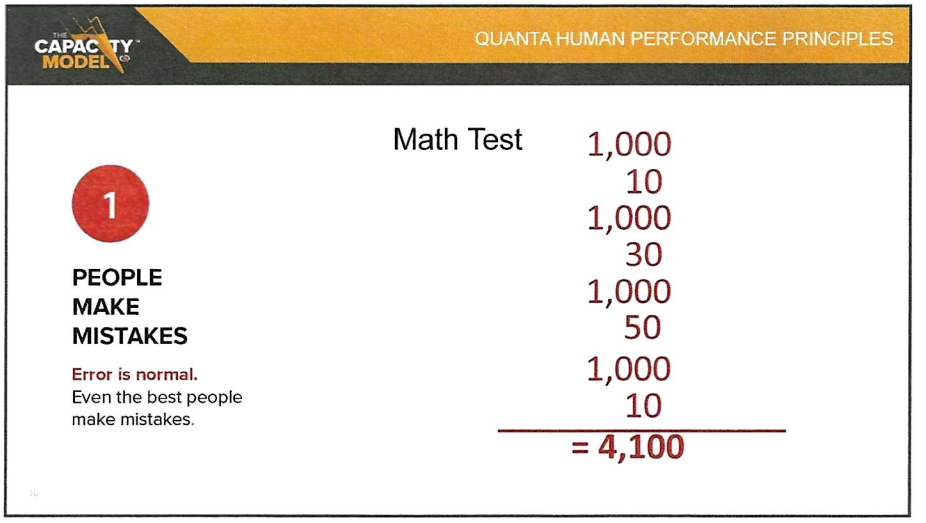
- We're going to solve a math problem. Your job is to give the total sum of the numbers.
- As I show the numbers, do the math in your head.
- When I say "go" at the end of the calculation, you will say the total sum aloud.

For slides 21-28: Give trainees three seconds to add the new number on the slide before advancing to the next slide.

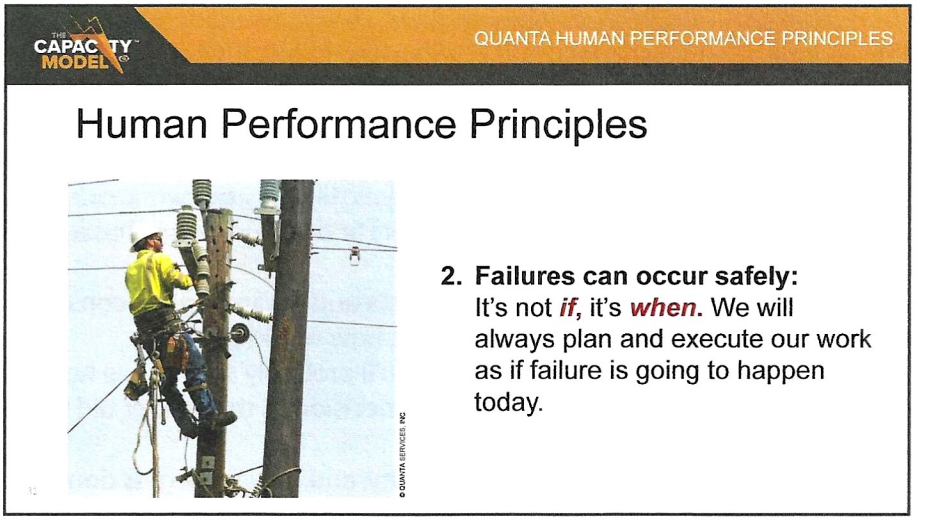


- When the final number is displayed, give trainees three seconds, and say "GO!" to say the answer aloud.

- Expect the answer 5000.
- The correct answer is 4100.
- Explain that as a function of how the human brain works, the timing and the pattern of numbers usually produce an error when adding the numbers.
- This is a very simple example of how people make mistakes.
- The question is: "Do we regularly ask our crews to do tasks more difficult or complex than adding eight numbers together?"



- Reiterate Principle 1: "People make mistakes. Error is normal. Even the best people make mistakes."

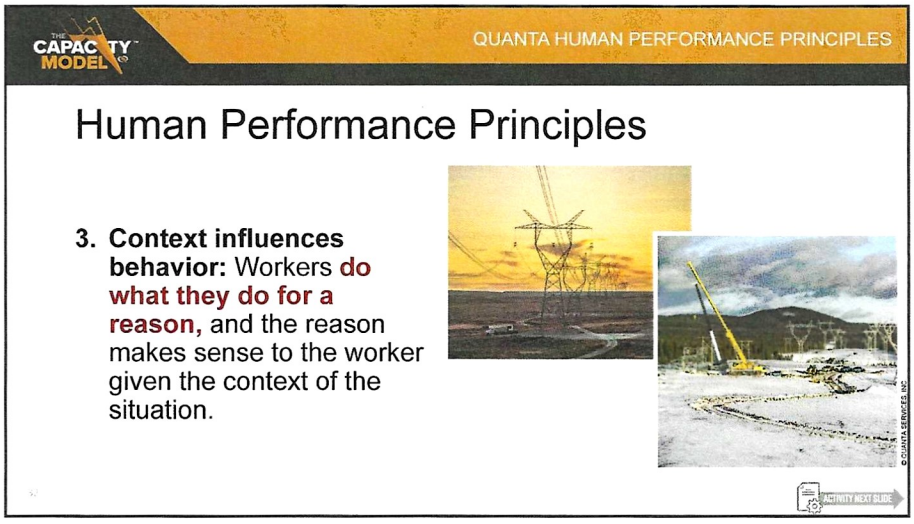


Slide aligns with page 18 of the trainee manual.

- Read #2 aloud.

PRESENT THE FOLLOWING:

- Humans are remarkably adaptable. With the complexity of our work, we often must adjust how we plan to do the work.
- Our senses, intuition, and training help us detect danger in real time and adjust.
- These internal warning bells are like the alerts in your car that warn you when you drift out of your lane or follow too closely to the car in front of you. For most errors and failures, we can adapt to the situation and continue without losing a beat.
- We cannot, however, reliably and consistently predict all future mistakes or events.
- As a result, we must prepare for when things go wrong, not if things go wrong, especially in those critical work tasks that we cannot live with the consequence of a failure.
- We must start thinking ahead and consider reasonable scenarios that could occur and what controls would be necessary to keep people from getting hurt when they do happen.

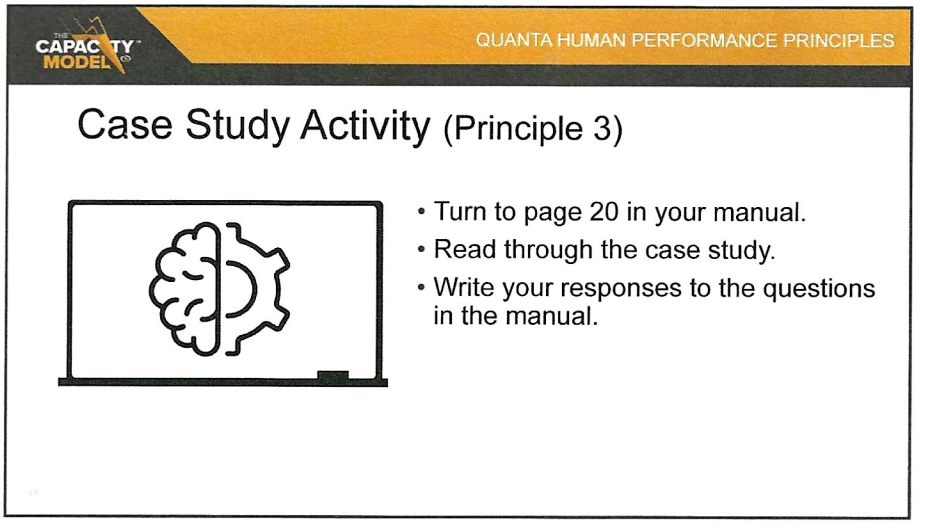


Slide aligns with page 19 of the trainee manual.

- Read #3 aloud.

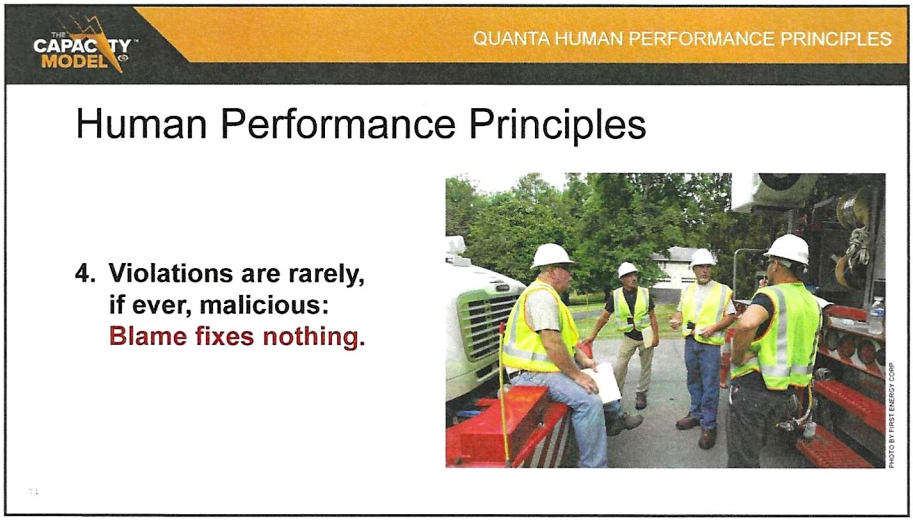
PRESENT THE FOLLOWING:

- When we look at errors and failure, it is important to examine them from the worker's perspective as the work unfolds.
- Examining things from the worker's perspective is very different from how a typical incident report or investigation looks at a failure: a linear description of A to B to C.
- As we all know, this linear description is not typically how our work, our lives, or the mistakes we make occur. The situation workers find themselves in has a multitude of variables and scenarios.
- This is called context. Understanding context is understanding the consistent variations that require us to frequently adjust how we do our work.
- If you look at the context around a failure, you'll probably start to see how you could have made the same error, mistake, or decision as the worker did if you had been in that situation in their place.
- Context tells the story of complexity, variability, and how work gets done.



ACTIVITY:

- Direct trainees to page 20 to read the case study and write their answers to the three questions.
  1. Have you experienced or seen "swift investigations" that ended with "human error" as the cause? Give an example.
  2. What did the company learn by suspending the two supervisors?
  3. How effective is it to suspend or replace workers if the system stays the same?
- Walk around the room to provide support.
- Ask for volunteers to share their responses.
- Discuss responses as a group.



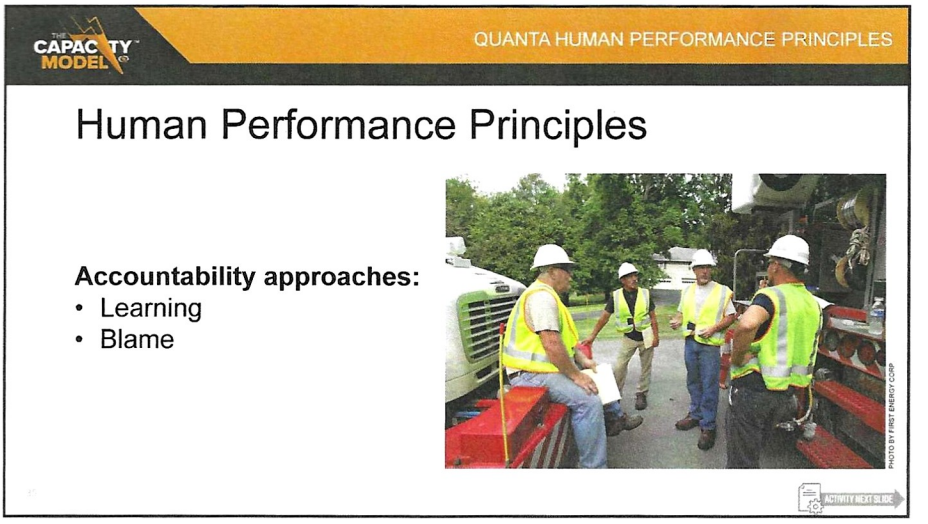
Slide aligns with page 23 of the trainee manual.

- Read #4 aloud.

PRESENT THE FOLLOWING:

- When it comes to errors or failure, it is extremely rare to see willful, criminal, sabotage-type behavior with the intent to injure, endanger, or destroy.
- If the failure or mistake was not premeditated, and we know that people make mistakes, then blame can only hurt, not help.
- Blame might feel productive at the moment, but it does not fix a complex problem.
- It serves no real or useful purpose and makes it much harder to discover and study future near-miss events.
- Nobody will want to report a near-miss if they think they might get blamed.
- Without those near-misses to look at, you cannot learn from them.
- Blame leads us away from learning after an event takes place.

Reinforce the guiding question:
How can our responses to an outcome effect change?



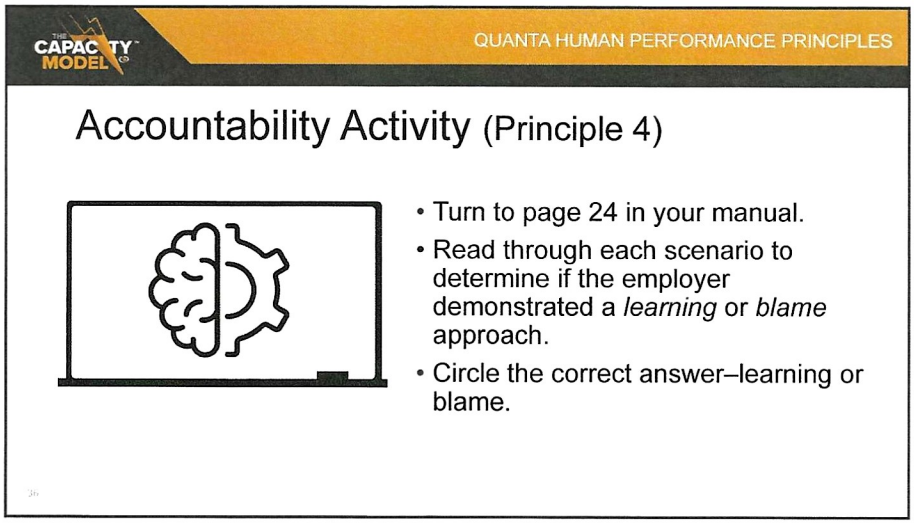
- Direct trainees to page 23, the last paragraph.
- Read the definition of accountability.

PRESENT THE FOLLOWING:

Explain the two different approaches to accountability:

- **The learning approach** embraces principle four and holds people accountable for their actions in a way that promotes learning.
- This approach looks at processes to see if improving those would prevent or mitigate a similar incident in the future.
- The learning approach seeks to understand the work.
- The goal of the learning approach to accountability is to improve systems and processes.

- **The blame approach** goes against principle four and holds people accountable for their actions by blaming them for an incident.
- The blame approach seeks to punish employees to make sure the mistake won't happen again.
- This approach looks at ways to fix employees as if they are broken.
- The goal of the blame approach to accountability is to find fault and punish.

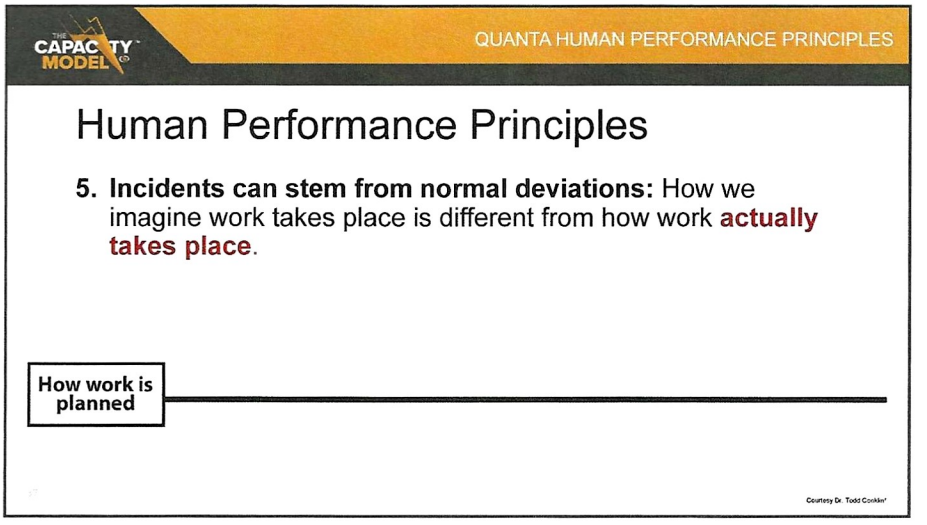


ACTIVITY:

- Direct trainees to page 24 in their textbook to complete the activity.
- Read through the instructions.
- Complete the first scenario as a class.
- Give trainees 5-10 minutes to categorize the remaining scenarios.
- Once the trainees have completed the activity, go through each scenario and ask for volunteers to offer their answers.
- Ask trainees to explain why they chose their answer.
- If trainees are answering incorrectly, ask leading questions, such as "Does this scenario point to blame?" Help the trainees correct themselves.

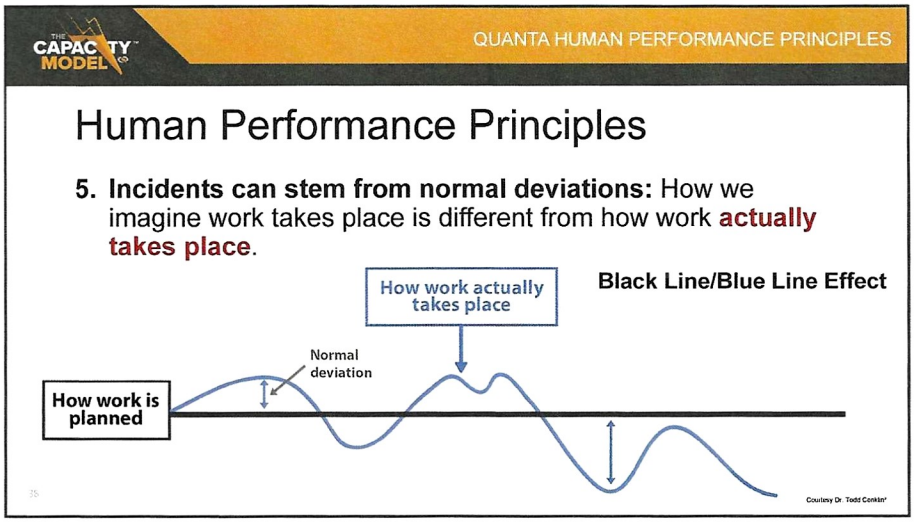
Accountability Scenario answers:

1. Blame
2. Learning
3. Blame
4. Learning
5. Blame
6. Learning
7. Blame
8. Blame
9. Learning
10. Learning



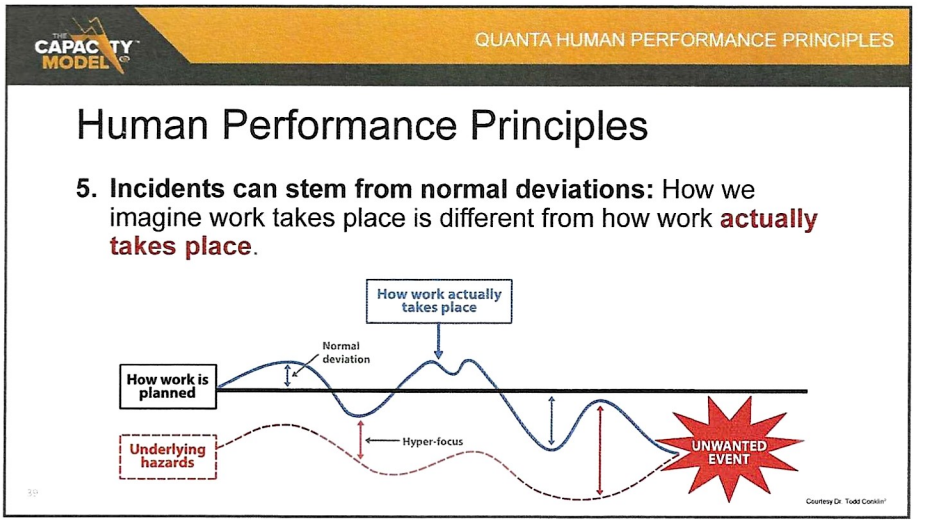
Slide aligns with page 26 of the trainee manual.

- Read #5 aloud.
- Ask trainees: Why do you think work takes place differently than how it's planned?



PRESENT THE FOLLOWING:

- The complexity of work, as we talked about in Principle 3, "context influences behavior," requires workers to adjust on an ongoing basis, so they can safely accomplish what they have been asked to do.
- We need to think of the work being completed–the blue line–as uncertain, complex, and ambiguous. This is the reality of work, and we need to help our teams account for it to make sure they maintain that margin of error to fail safely.
- All work inevitably drifts away from the perfect procedure, practice, and plan.
- We call this the black-line/blue-line effect: Work, as planned or imagined, is the black line, and work, as executed, is the blue line.



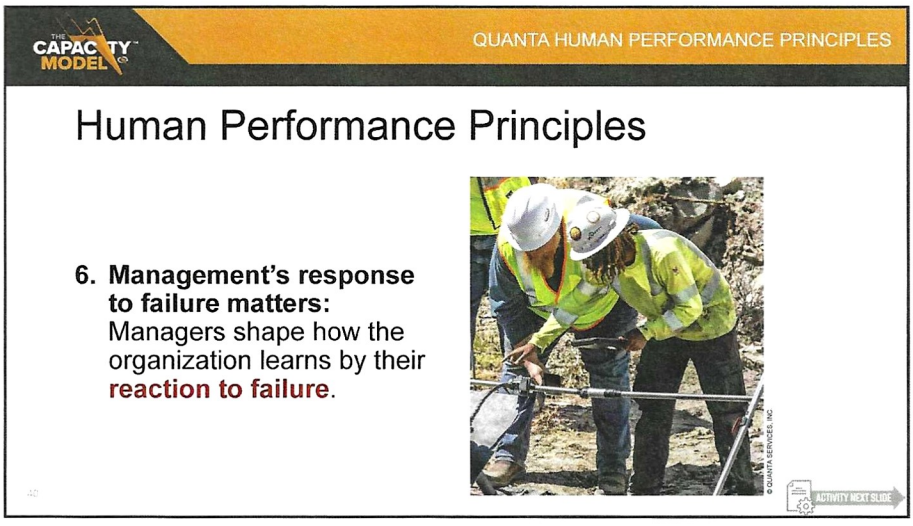
- Ask trainees: Do you have any examples you can share of when the work you did was completed differently than how it was planned?
- Call on 2-3 volunteers to share their stories.
- As trainees share their stories, reinforce how the work was planned versus how the work was executed and why it was executed differently.

PRESENT THE FOLLOWING:

- The red line represents real-world conditions that can be hazardous. If the blue line drifts too far and connects with the red line, an unwanted event can occur.

Provide the following example (or use one of your own):

- Imagine the black line is your daily commute to work. You know it well; in fact, you are the expert on getting from home to work.
- As a car owner, you probably have mitigative controls in your vehicle: jumper cables, spare tire, tire changing kit, and so forth.
- Before you start your commute, you implement preventive and mitigative controls: adjust your mirrors, put on your seatbelt, and place your phone on "Do not disturb."
- The blue line represents an unexpected deviation in your commute. Maybe a road you normally take is closed. Maybe you need to stop at a gas station or other location that is not on your normal route.
- On the alternate route to work, you encounter a hazard (red line)–a pothole in the road. You do not see the pothole in time and drive into it, causing an unwanted event–you blow a tire.
- Because you have mitigative controls in place–a spare tire and tire changing kit in your vehicle–you built in capacity for failure.
- The potential for unwanted events is everywhere, and we can do our best to avoid them.
- The only way to ensure we can withstand these events is to put controls in place so that when they occur, we can absorb them safely.

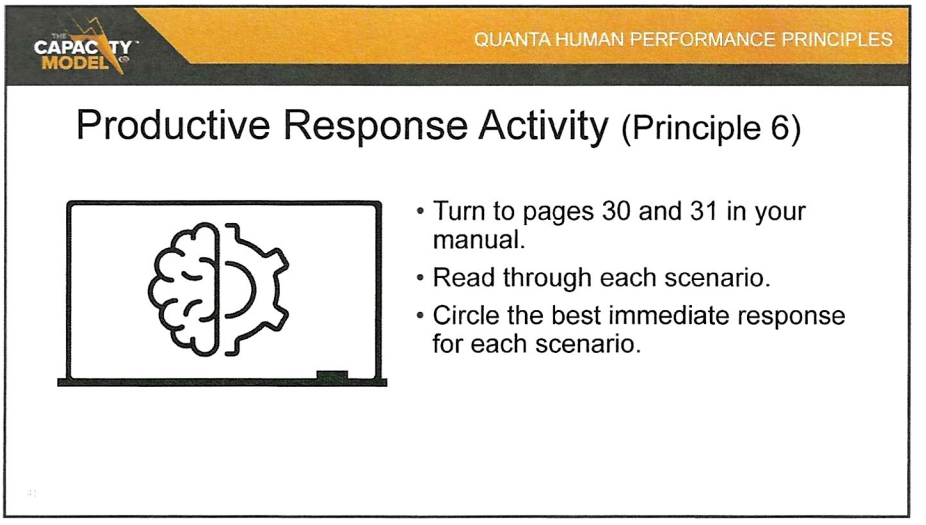


- Read #6 aloud.
- Direct trainees to page 27 in the manual.
- Review the difference between an unproductive reaction and a productive response.
- Use Table 1.1 on page 29 and walk through the unproductive reactions and productive responses.
- Ask trainees: Give an example of a time when a leader had a productive response or an unproductive reaction to a mistake or failure. How did that affect you and your crew?

PRESENT THE FOLLOWING:

- If we accept that people make mistakes, our work procedures and systems can bring about mistakes, and mistakes are not malicious, then how management responds to failure does matter—a lot.
- As leaders, everything we do sets a standard for behavior.
- How we respond to success or failure sets the framework for relationships, credibility, respect, trust, and future results.
- How you treat failure can strengthen or break apart that framework.
- Failure is common, so we need to think constructively about how we respond to someone who has made a mistake and failed.
- If we treat it as an opportunity to learn and improve, we open channels that give us insight into our systems and strengthen our capacity to fail safely.
- Remember, context matters, and blame moves us away from learning.

Reinforce the guiding question:
How can our responses to an outcome effect change?

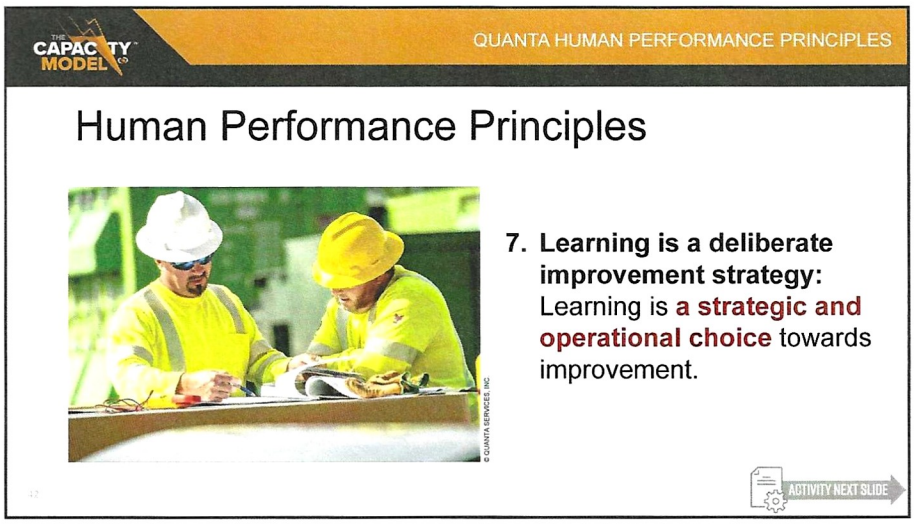


ACTIVITY:

- Direct trainees to pages 30-31 in their manual to complete the activity.
- Reinforce the instructions: After an unwanted event, you will likely ask many of the possible questions. In this activity, we are focusing on the first response.
- Once the trainees have selected their answers, ask for volunteers to share their responses.
- Ask them to explain why they chose the response.
- If trainees are answering incorrectly, refer them back to the table on page 29 of the textbook and ask the trainees leading questions such as, "Does the answer you chose take time, require thought, help calm, and invite a discussion?" or "Does the initial response focus on the safety of the people?"
- Help the trainees correct themselves.

Productive Response Answers:

1. A
2. C
3. B
4. D

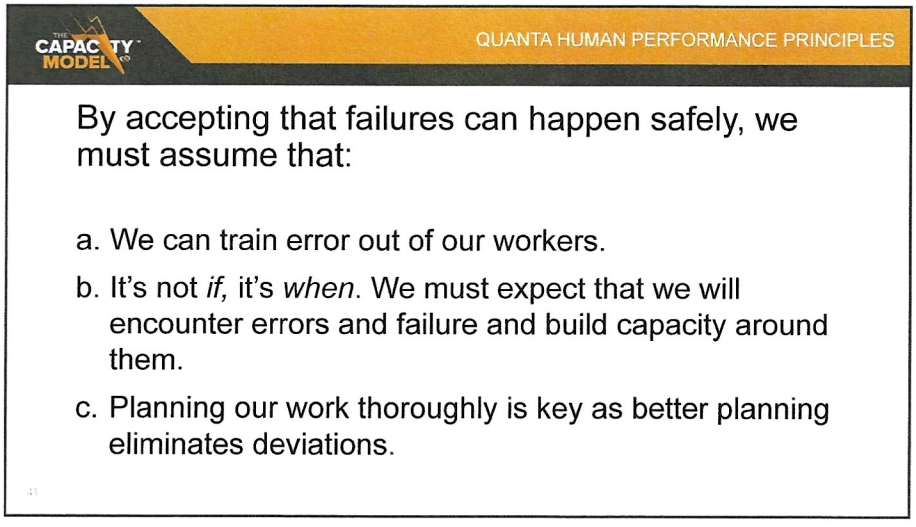


Slide aligns with page 32 of the trainee manual.

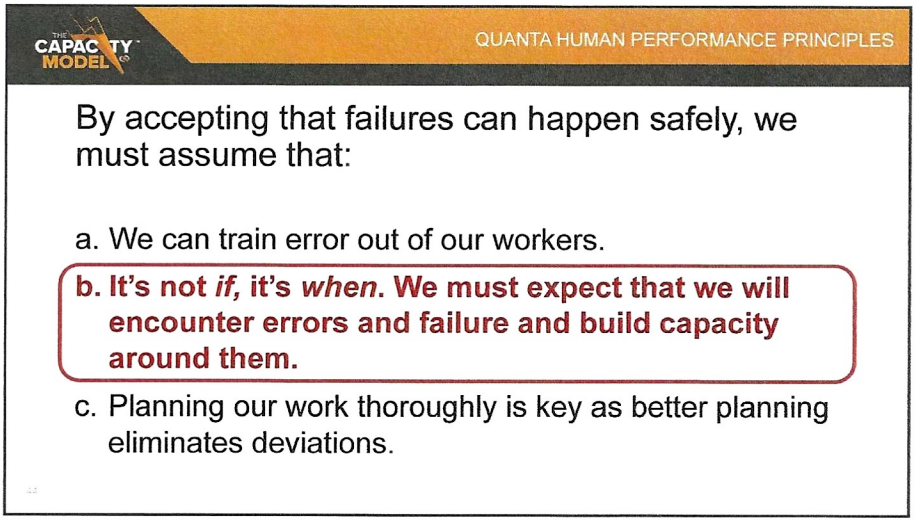
- Read #7 aloud.
- Ask trainees: Why should leaders pay attention to the successes as well as the failures?
- Give trainees 1-2 minutes to think about their responses.
- Call on 2-3 volunteers to share.
- Validate answers.

PRESENT THE FOLLOWING:

- Company leaders who make an effort to learn and consistently demonstrate productive responses to successes and failures will change the organization in a positive way.
- You must study success and failure to keep your organization sharp and strengthen everyone's capacity to fail safely.
- When we choose to learn from incidents, near misses, patterns, trends, and successful work, we can share our learnings with the rest of the organization.
- We must remember that error is normal, so if everything is "perfect," we either need to learn the source of that success or uncover the hidden problems.
- Sometimes, "no problems" can be hiding a lot of serious, challenging issues that can build, transform, and quickly turn into something catastrophic.
- Because "when nothing is happening, everything is happening." What does this mean?
- When it is quiet, and you are not seeing a lot of incidents, that does not mean you are safe. You must ask the question, "Are we good, or are we lucky? How do we know?"
- There is a lot happening to ensure that things are going well. We need to understand why.
- Safety is not the absence of unwanted events. It is the presence of capacity to safely absorb unwanted events.
- Perfect work, the perfect day, and perfect results are bright signals that deserve attention. We must look at these situations as opportunities to learn and improve.



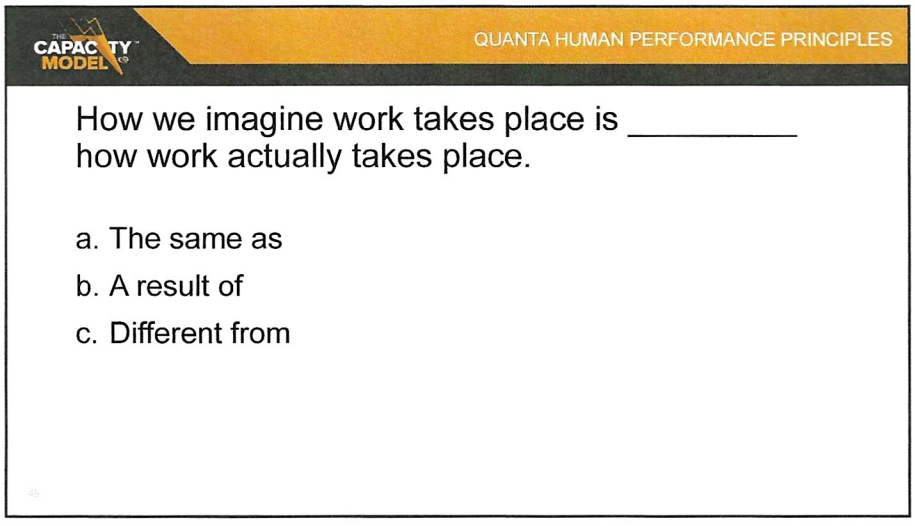
- Read the question slide aloud to trainees.
- Give them time to write their answer or read the answers aloud and have them raise their hands for the one they think is correct.



- b is the correct answer.

PRESENT THE FOLLOWING:

- We must assume that errors and failures will occur. Only then can we build capacity to absorb them.
- We must understand that we can't train human error away.
- And planning our work, while good, can't possibly predict all normal deviations.



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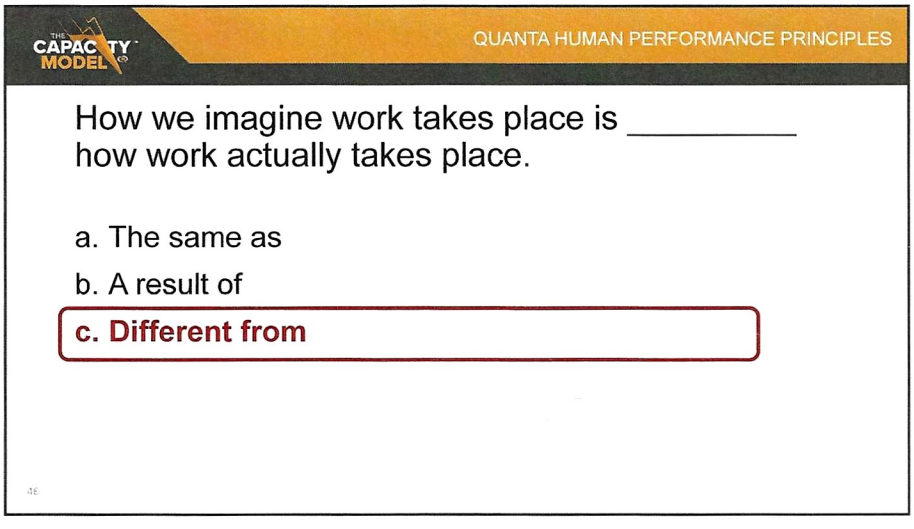
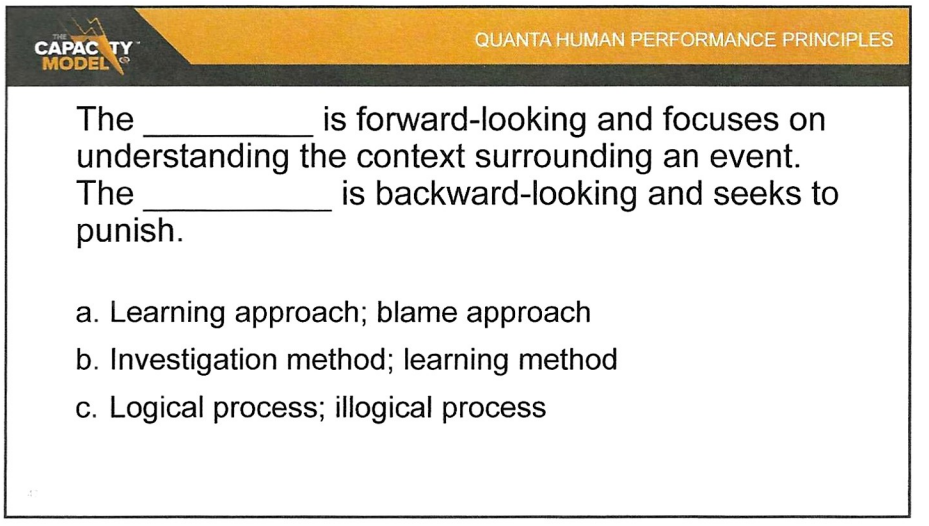
- Give them time to write their answer or read the answers aloud and have them raise their hands for the one they think is correct.



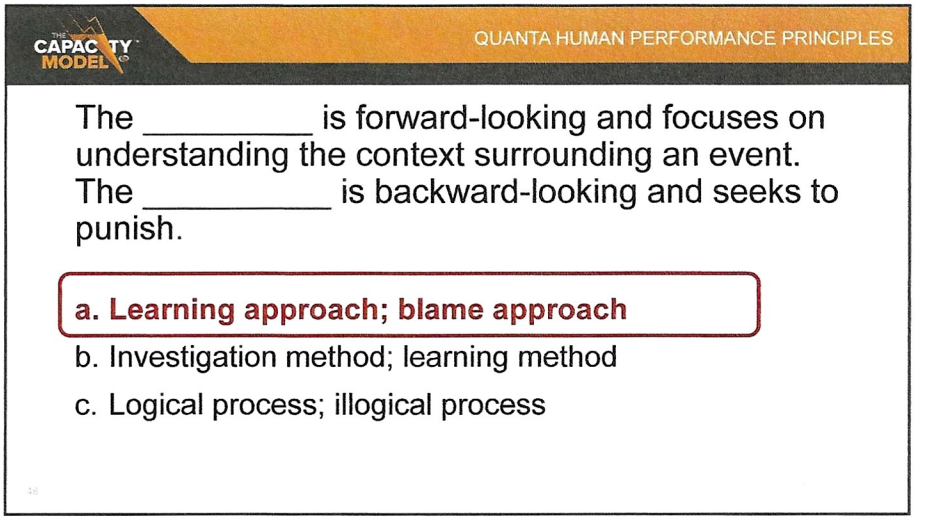
- c is the correct answer.

PRESENT THE FOLLOWING:

- Work virtually never takes place exactly how we plan it.
- Planning work is necessary, but it won't protect us from all potential errors or failures.



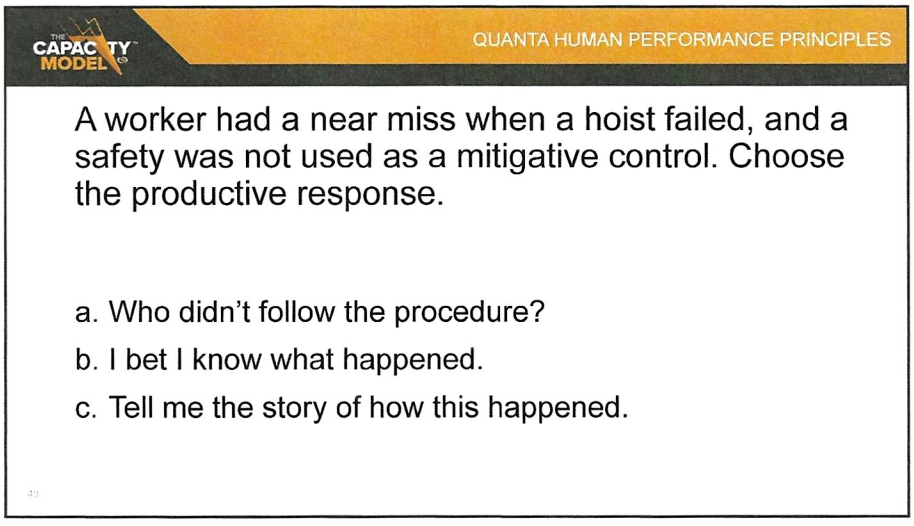
- Read the question slide aloud to trainees.
- Give them time to write their answer or read the answers aloud and have them raise their hands for the one they think is correct.



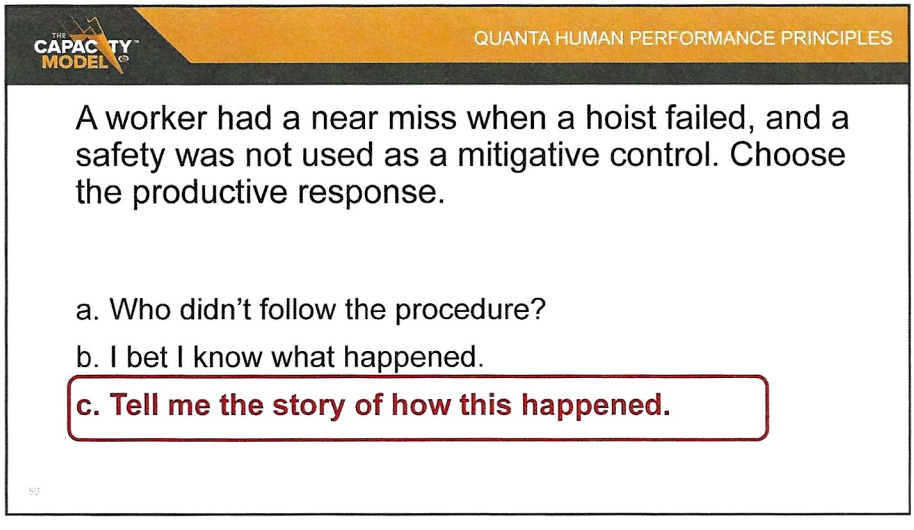
- a is the correct answer.

PRESENT THE FOLLOWING:

- The learning approach seeks to learn from an event and take steps to improve.
- The blame approach seeks only to find who to blame and punish them.
- With the blame approach, failure will likely occur again because learning and improvement are absent from the process.



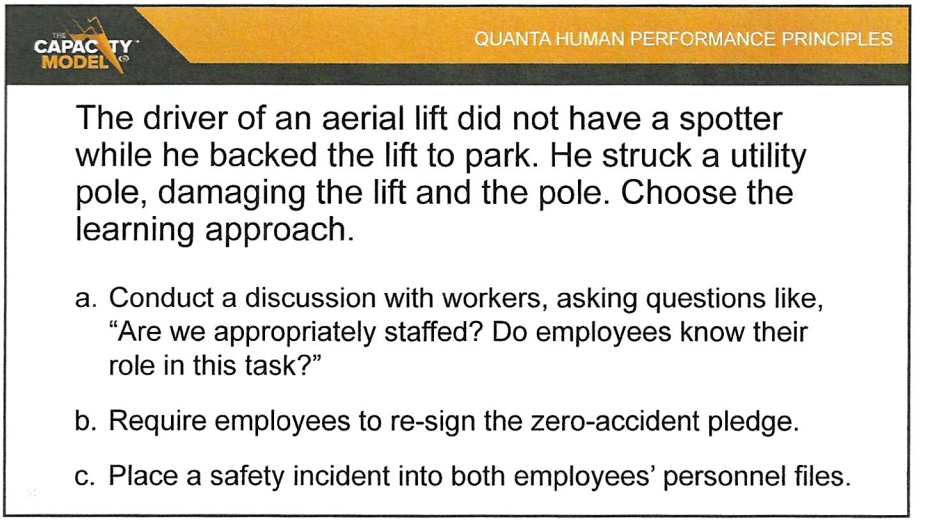
- Read the question slide aloud to trainees.
- Give them time to write their answer or read the answers aloud and have them raise their hands for the one they think is correct.



- c is the correct answer.

PRESENT THE FOLLOWING:

- The unproductive reactions jump to find blame and do not allow anyone to learn from the event.
- The productive response allows the workers to discuss the event so that learning and improvement can happen.



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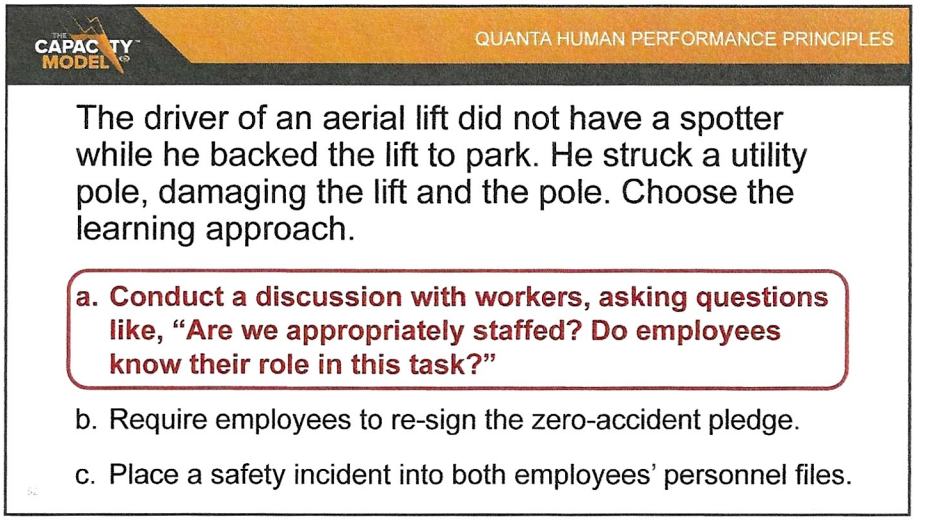
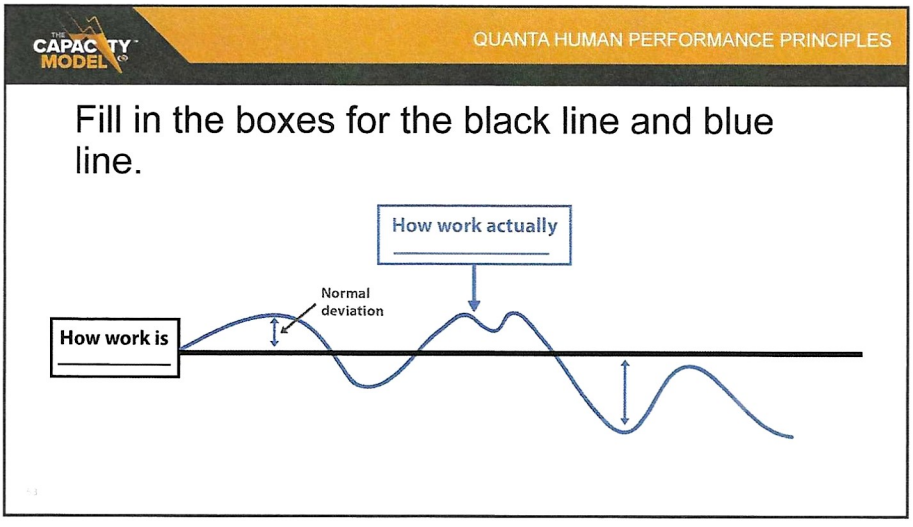


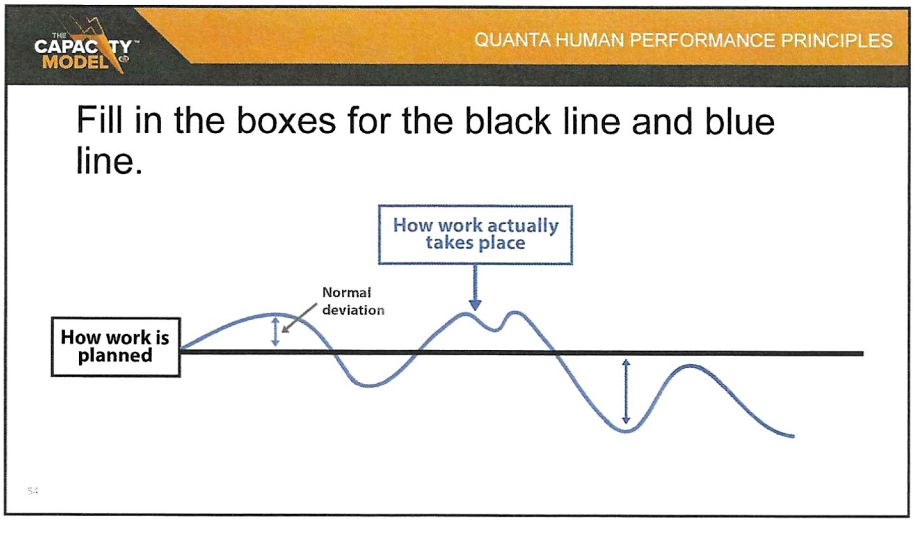
- a is the correct answer.

PRESENT THE FOLLOWING:

- Option b is a punishment and does not allow the workers to learn from failures.
- While option c may be necessary, only choosing this option will not help to prevent future failures.
- Option a involves the crew, allowing them to learn and improve.



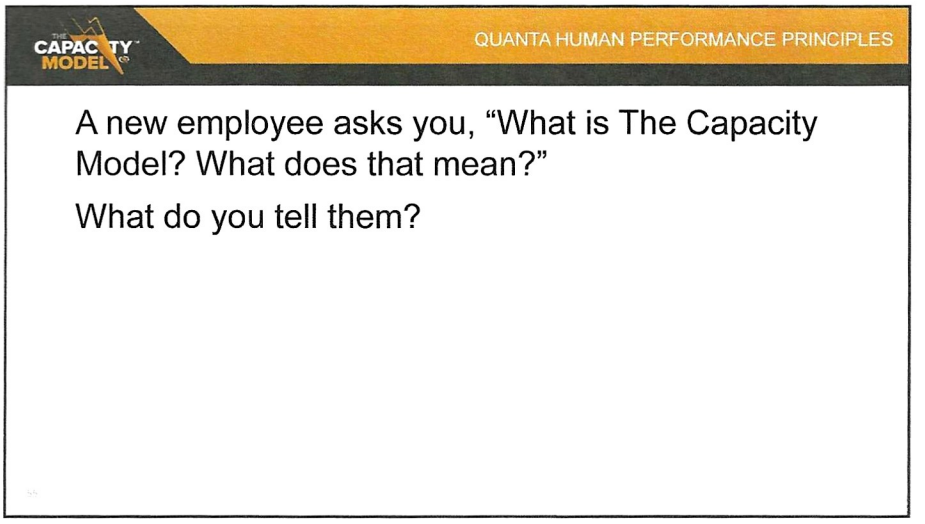
- Read the question slide aloud to trainees.
- Give them time to write their answer.
- Ask for volunteers to share their responses.



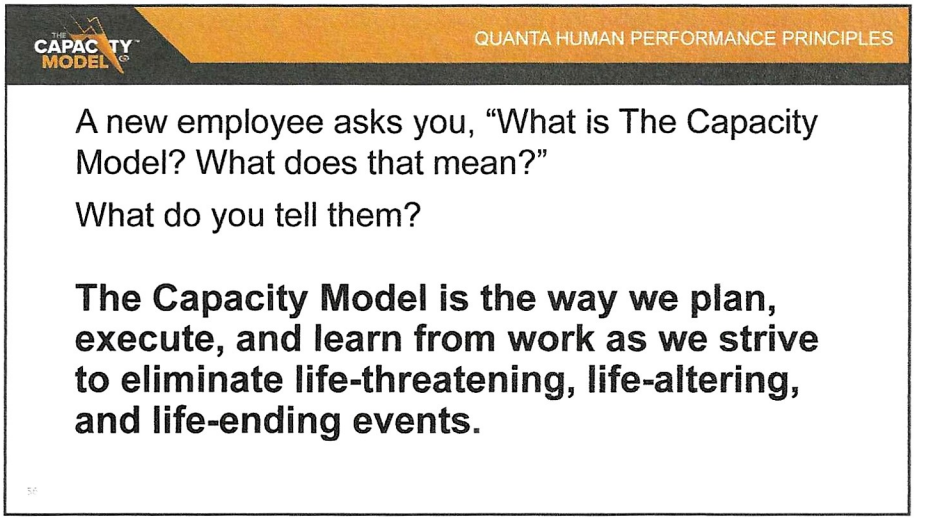
- Black line: How work is planned.
- Blue line: How work actually takes place.

PRESENT THE FOLLOWING:

- Work is planned based on standard methods for completing a task, such as installing a utility pole–black line.
- Ground conditions, weather, terrain, and environment will affect how that work is completed–blue line.
- How a utility pole is installed in a small rural town is much different than installing a utility pole in a large city.
- *Ask trainees:* What does the red line represent?
- *Confirm the red line represents underlying hazards.*



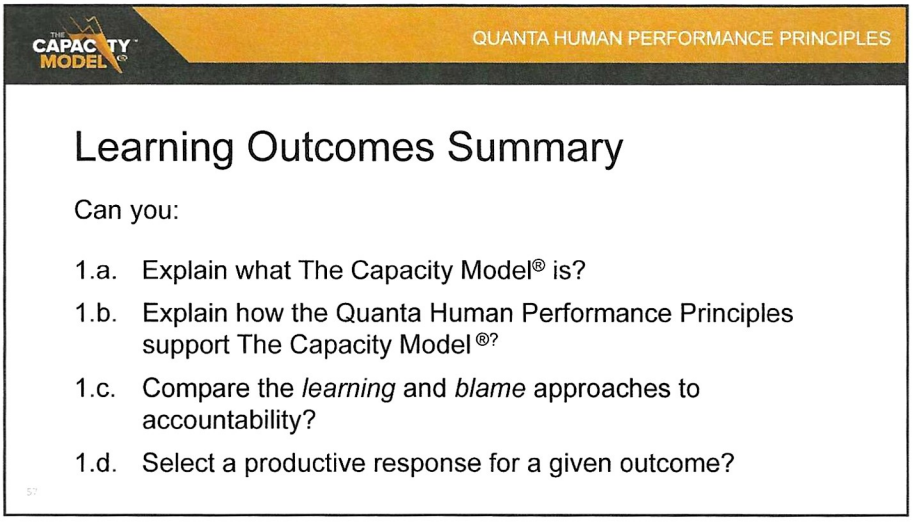
- Read the question slide aloud to trainees.
- Give them time to write their answer.
- Ask for volunteers to share their responses.



PRESENT THE FOLLOWING:

Reinforce the meaning of The Capacity Model:

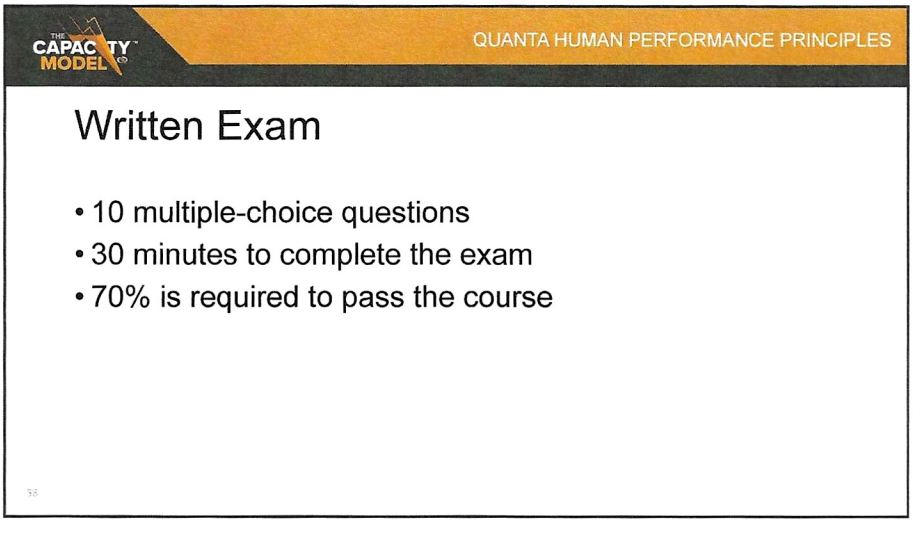
- Eliminate life-threatening, life-altering, and life-ending events: While we always want to eliminate and mitigate all events that can cause injuries, The Capacity Model focuses on STKY events – the stuff that kills you.
- Plan work: Everything we do before the work begins allows us to focus on the tasks we perform and the preventive controls that are necessary.
- Execute work: But remember, prevention is not enough. As people execute the work in real time, they are continuously adapting and adjusting to normal conditions and variability of the job. We know that sometimes, things do go wrong, so we will always plan our work as if failure will happen today and build in the capacity to fail safely when it does.
- Learn from work: Whether learning from when things go wrong, or from things that go right, we know that learning is a deliberate improvement strategy. The only way to get better is to take the time to learn from those who do the work.



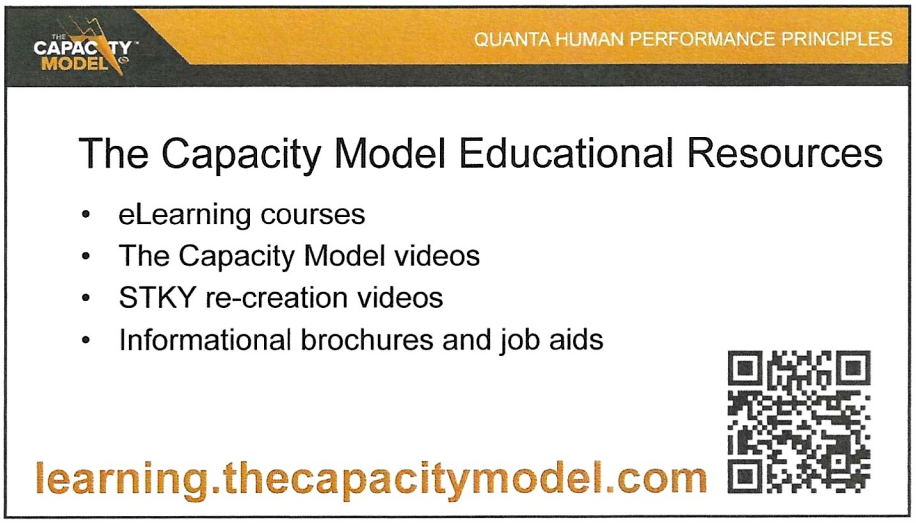
- Review the learning outcomes. Discuss each with the trainees. For example:
  - Ask for volunteers to summarize learning outcome A.
  - Ask volunteers to share one thing they learned related to one of the human performance principles.
  - Give an example of a learning approach and blame approach. Ask for volunteers to identify the approach and explain why.
- Clarify anything trainees do not understand.
  - Ask trainees: What are your questions?

After asking for questions, count to seven in your head before moving on.

See Appendix D for other Formative Assessment ideas.



Review bullet items for the written exam.



- Direct trainees to learning.thecapacitymodel.com or the QR code to view and download TCM educational resources.
- Explain the site is open. They do not need to register or log in.
- Review the resources:
  - eLearning courses such as STKY Controls for Crews and Learning Teams.
  - TCM videos like those viewed in this course.
  - STKY re-creation videos are re-creations of real STKY incidents from Quanta operating companies.
  - Informational brochures and job aids such as the STKY discussion questions are available to download.

EXAM GUIDE

- Pass the exams out to the trainees.
- After everyone finishes, instruct trainees to swap exams with a partner.
- Review and grade the exams as a class. Discuss each question and answer.
- Collect exams.

WRAP-UP

- Ask trainees to complete the appropriate instructor evaluation form.

APPENDICES

APPENDIX A—COURSE DELIVERY TIPS

BEFORE DELIVERY

- Review the instructor manual to stay fresh on the information.
- Star your favorite application activity and formative assessment ideas in the appendix so you can use them in your lesson.
- Make sure you have all your materials ready and accessible.
- Set the room up in a way that encourages discussion and participation in activities. For example, having the students sit around tables instead of in rows.
- Use the tools available to you. Don't have access to mini whiteboards? What about a page protector with a piece of printer paper in it?

DURING DELIVERY

- Build a positive, professional classroom culture.
  - Set clear expectations.
  - Encourage all questions and comments – this is a safe place. Disagreements can be made in a professional and respectful way.
  - Complimenting can go a long way.
  - Smile!
  - Provide breaks and monitor student body language to decipher when to use them. Consider designating a student to help you know when it is a good time for a break.
- Don't just read the learning outcomes – decipher them for the student. What is in it for them?
  - Write the learning outcomes on the whiteboard or on a wall chart to reference throughout the lesson.
- Start your lessons by motivating the students and re-motivate throughout your delivery.
  - A quick and easy way to monitor engagement is to walk around the room during the lesson.
  - Making eye contact is an excellent way to motivate students to stay engaged.
- Spend plenty of time on the application activities—these are the meat and potatoes of the lesson!
- Plan three to five extra application activities to use in the lesson. If you have extra time incorporate them. If not, it's no big deal!
- Involve students in the presentation – you should be talking less than they are!
  - Use students' experiences to help each other learn. Your job is to facilitate and guide the discussions around the topics.
- Evaluate throughout the lesson to check student understanding before moving on to a new concept.

AFTER DELIVERY

- Write notes in your instructor manual to improve your delivery after each lesson.
  - Make note of things you would like to try.
  - Make note of things that went well.
  - Make note of things that were challenging.

APPENDIX B—RECOMMENDED RESOURCES AND MATERIALS FOR ACTIVITIES AND FORMATIVE ASSESSMENTS

Each activity or assessment in Appendices B, C, and D has accompanying materials and/or other components to prepare before class starts.

The following is a list of recommended materials to have on hand so you can be prepared for most of the activities. You should make sure you have enough of each item for each student to have a set.

NAME	DESCRIPTION
ABC Cards	All letters of the alphabet printed on cardstock and cut out. *Tip: Single-hole punch a corner and stick them on a binder ring so that they don't get lost or mixed up easily.*
Dry erase markers	Thin tip dry erase markers in assorted colors.
Chart Paper	Large sheets of unlined paper that can be attached to the wall by either adhesive on the back or tape.
Envelopes	Envelopes for activities that require sorting or containing small cutout items.
Lined Paper	College ruled or wide ruled lined paper
Magnets	A baggy of small magnets to use with magnetic whiteboards.
Markers	Markers in assorted colors.
Pens/Pencils	Black or blue pens and number two pencils.
Printer Paper	White unlined paper.
Tape	Masking or painters tape so that it will stick to most surfaces without damaging anything
Whiteboard erasers	Individual small whiteboard erasers, or new socks or wash cloths for increased portability
Whiteboards	Individual student whiteboards, or a piece of printer paper in a page protector for increased portability.

APPENDIX C—APPLICATION ACTIVITY MENU

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
3-2-1	1. Ask students to list three things. 2. Ask students to list two things. 3. Ask students to list one thing. *Example: Write the three elements of The Capacity Model™. Write two scenarios depicting STKYa events. Write one question you still have about the topic.*	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
ABC Words/Phrases	1. Give each student, pair, or group a letter. 2. Have students find a word or phrase from the lesson that starts with that letter. 3. Have students share their words/phrases.	*Optional:* • ABC Cards
Acrostic	1. Have students write a lesson-related word vertically. 2. Have them use each letter in the word to write a word/phrase that relates to the topic. 3. Have students volunteer to share.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Baseball	1. Have students divide into two evenly-sized groups. 2. Have two players from each group come up to the whiteboard. 3. Provide a question. 4. The first player to write the correct answer gets a point for their team. 5. Those two players go back to their team and two other players come up to the whiteboard. 6. Repeat the process as many times as needed. 7. The team with the most points at the end wins. You choose the prize.	• Wall-mounted whiteboard ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers ***OR*** • Paper • Pens
Chain Writing	1. Give the class a topic to write about. 2. Pass around a piece of lined paper. 3. A student writes one sentence about the topic and then passes it on to the next student. 4. Continue until the topic is completely written about. *Note: Every few students, glance at the paper to see if they have written enough.* 5. Read what the students wrote aloud to the class.	• Lined paper • Pens
Elevator Pitch	1. Have students prepare a 1-2 minute speech summarizing the current topic. 2. Have students give their speeches to the class.	*Optional:* • Paper • Pens
Envelope Sort	1. Provide categories for students and envelopes that have passages/scenarios in them. Example: You could have envelopes for STKYa, STKYp, and STKY success. 2. Students work in partners or groups to sort the passages into the categories. 3. Discuss the correct answers with the class.	• Envelopes with passages or scenarios
Gallery Walk	1. Have students create a poster about a topic. 2. Put the posters up on the walls. 3. Have the class walk around and view each poster.	• Paper • Markers • Tape / Magnets

APPENDIX C—APPLICATION ACTIVITY MENU (cont.)

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
Graffiti Wall	1. Write questions or topics on posters or whiteboards around the room. 2. Have students rotate through and write their response to each question or topic. 3. Review the answers as a class.	• Chart paper • Pens ***OR*** • Wall-mounted whiteboard • Dry-erase markers • Whiteboard erasers
Jigsaw	1. Give each student or pair/group a letter. 2. Have students find a word or phrase from the lesson that starts with that letter. 3. Have students share their words/phrases.	• Paper • Pens • Reading passages *(if necessary)*
Line in the Sand	1. Have the class divide themselves evenly onto opposite sides of the room. 2. Explain that the middle of the room is the "line in the sand" and choosing a side to stand on indicates agreement with that side. *Example: The sides could be disagree/agree, or you could use opposing concepts, such as STKYp and STKY success.* 3. Label each side of the room. 4. Provide prompts to the students. 5. Tell them to stand on the side they think the prompt would fall under. 6. Provide the correct answer.	
My Favorite No	1. Using exit slips, a quiz, or another formative assessment, present your "favorite" wrong answer to the class. 2. Ask the students what is correct or done well first. 3. Ask students what the mistake(s) was. 4. Ask them how to correct it. *Note: You can find the description for exit slips in Appendix D.*	• Completed exit slips ***OR*** • Completed quiz ***OR*** • Another completed formative assessment sample
Pictionary	1. Put key words/phrases/topics that relate to the course and are drawable on slips of paper. 2. Have students take turns picking a slip of paper and drawing the word/phrase/topic on the whiteboard so the rest of the class can see. 3. The rest of the class guesses what the student is drawing.	• Key words/phrases/ topics written on slips of paper • Wall-mounted whiteboard ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers ***OR*** • Key words/phrases/ topics written on slips of paper • Paper • Pens

APPENDIX C—APPLICATION ACTIVITY MENU (cont.)

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
Quick Write	1. Say or put up a prompt. 2. Give students 3-5 minutes to write a quick response. 3. Ask for volunteers to share what they wrote.	• Paper • Pens ***OR*** • Individual whiteboards • Dry erase markers • Whiteboard erasers
Student-Created Questions	1. Have students create 1-3 questions from a selected reading or after a presentation. 2. Students take turns asking and answering each other's questions. 3. Review some of the questions as a class.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Table Discussions	1. Ask/put up a question. 2. Have students work in tables to discuss the question. 3. Have volunteers share their answers.	
T-Charts	1. Have students draw a large T on a piece of paper or their whiteboard. 2. Have students label the two sides of the T at the top. *Example: STKYp and STKY success.* 3. Have students provide examples under each column. 4. Have volunteers share their examples.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Think-Pair-Share	1. Ask/put up a question. 2. Have students think about their answer on their own, silently. 3. Have them share their answers with a partner. 4. Have students volunteer to share their answers with the class.	
Venn Diagram	1. Provide two topics for students to compare and contrast using a Venn diagram. 2. Have students volunteer to share their answers.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Who/What Am I?	1. Provide students with a key term, person, or topic. 2. Tell them they may not look at it and have to place it on their forehead. 3. Students must then go around asking their classmates questions that will help them guess who/what their slip of paper says. 4. When they think they know they can ask another student. 5. If the student guesses correctly, they can sit down. If they are not correct, they need to keep asking questions until they can guess the correct answer.	• Pre-made key terms, people, or topics on slips of paper
Word/Phrase Sort	1. Give students key terms or phrases written on cards. 2. Students sort the cards into categories. 3. Students share why they sorted the cards the way they did with the class.	• Pre-made key terms or phrases cards

APPENDIX D—FORMATIVE ASSESSMENT MENU

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
ABC Cards	Ask the students a question. Have students hold up a letter card to answer the question. These could be the A/B/C/D cards for multiple choice questions, or a specific letter that correlates to a topic you are working with. *Example: A worker fell from a bucket and wasn't wearing their harness, but didn't get hurt. Hold up an A if you think this would classify as a STKYa, P if you think this would classify as a STKYp, or S if you think this would classify as a STKY success.*	• ABC Cards
Circle the Questions	1. Provide students with a list of questions relating to the topic. 2. Tell them to circle the questions they can't confidently answer. 3. Review the circled questions to inform your instruction and support struggling students.	• Pre-made list of questions • Pens
Exit Slips	1. Pass out a simple, short question printed on a half sheet of paper at the end of class. *Note: The question doesn't have to be multiple choice—it can be open-ended, and it is often better that way.* 2. Give students 3-5 minutes to answer. 3. Collect the exit slips and review them.	• Pre-made questions printed on a half sheet of paper • Pens
Fingers Up	Have students hold up a number of fingers from one to five. One meaning "I need help!" and five meaning "I've got this and could teach others!"	
Hand Thermometer	Have students self-assess their learning by raising their hand to different levels. The closer to the ground the less they understand. The higher in the air the more they understand.	
Quick Nod	These are great for yes or no type questions. Ask a question and students either shake or nod their heads.	
Quiz	Give students a quiz relating to the topics. Review their answers to inform your instruction and support struggling students.	• Instructor-made quizzes ***OR*** • Course quizzes • Pens
Summaries	Have students summarize, in their own words, what they learned during that day/segment. Review these answers to inform your instruction and support struggling students.	• Paper • Pens
Thumbs Up / Thumbs Down	To have students self-assess their understanding, you can ask them to show you thumbs up for if they've got it, thumbs down if they are confused, or even thumbs to the side if they understand a little but could still use some help.	
Traffic Light	Have students use a traffic light format to convey their understanding. Red light for completely lost and no understanding; yellow light for some understanding but needs additional support; green light for complete understanding—no support needed.	• Paper • Red, yellow, and green markers ***OR*** • Individual whiteboards • Red, yellow, and green dry-erase markers • Whiteboard erasers

APPENDIX E—EXTENSION ACTIVITY MENU

These activities are meant to be used for students who are excelling and need extra practice, or if you have extra time in the class and need to fill it in a meaningful way.

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
ABC Books	Have students create an alphabet book with each letter representing something from the concepts taught	• Paper • Markers • Pens
Comic Strips	Have students create a comic strip depicting a concept from the course.	• Paper • Markers • Pens
Create the Test	Have students create a test with a key. It can be multiple choice or have open-ended questions.	• Paper • Pens
One-Pager	Have students create a one-page document summarizing concepts taught. Use unlined paper for this activity. Students should have the following on their page: • Title • Illustration • Quote • Unlined paper • Questions & answers *(minimum of 2)*	• Unlined paper • Markers • Pens
Posters	Have students create posters summarizing the concepts taught.	• Paper • Markers

APPENDIX F—ONLINE LEARNING FOR THE CAPACITY MODEL

The Online Learning for The Capacity Model website provides valuable educational resources. The site is open to everyone. There is no registration requirement. Instructors can use content to supplement their training or share the site with trainees, who can use it as a resource to refresh and extend their learning.

ACCESS THE WEBSITE

Go to: https://learning.thecapacitymodel.com.

ELEARNING COURSES

Courses are available in English, Spanish, and French. Options include:

- Understanding The Capacity Model
- STKY Controls for Crews
- Introduction to Learning Teams

TCM VIDEOS

The videos provide viewers an introduction to several elements of The Capacity Model. Some of the options include Quanta Human Performance Principles, STKY, Quanta Energy Wheel, and Quanta's Safety Philosophy.

STKY RE-CREATION VIDEOS

The re-creation videos are based on actual events. Instructors can use these videos to supplement STKY and the Energy Wheel training to reinforce concepts such as using the Energy Wheel, asking capacity questions, identifying preventive and mitigative controls, and classifying STKY events.

RESOURCES

The site's Resources section includes PDFs of printable informational brochures and job aids such as the STKY Discussion Guide.

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